ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2022)4648166 - 24/06/2022

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101070666
Project name:	Up-Skilling for Industry 5.0 Roll-Out
Project acronym:	Up-Skill
Call:	HORIZON-CL4-2021-HUMAN-01
Topic:	HORIZON-CL4-2021-HUMAN-01-26
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 September 2022
Project duration:	36 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

The 4th Industrial Revolution/Industry 4.0 has enabled reduction of production costs, improved consistency of product quality and enabled mass customisation by merging the physical and digital worlds. The transition is still ongoing Industry 4.0 is a general-purpose technology, adding value across all industrial sectors. However, the perception of Industry 4.0 at a human level has not all been positive. It has been plagued by fear of job cuts and in some sectors completely replacing the human workforce. Automation projects have often failed due to omitting the critical skilled human elements in business success with unintended consequences including reduced customer satisfaction, poorer product quality and lower process efficiency. Automation alone clearly cannot be a source of sustained competitive advantage. I5.0 will address the balance between humans and technology, focussing on the collaborative relationship between skilled workers and automation. The intent is reinstate skilled craftsmanship at the centre of production processes where people add unique value and competitive advantage, augmented by intelligent, data-driven technology emerging from Industry 4.0.

In the Up-Skill project, we will address the implications of Industry 5.0, in particular the relationship between automation, skilled work and organisational systems. Our research will establish how the relationship between automation and human input plays out in a range of industrial settings, creating comparative case studies to capture effective implementation strategies. We will address under-explored strategic spaces in production - where automation adds value to skilled and artisanal work, and where further automation risks undermining product value. This research will identify the shifting organisational characteristics that are needed to ensure technology advancements are implemented within companies while ensuring sustainable, added value for man, machine, and organisation.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	MDU	MALARDALENS UNIVERSITET	SE	999881530
2	BEN	MITC	MALARDALEN INDUSTRIAL TECHNOLOGY CENTER AB	SE	911914170
3	BEN	UMIL	UNIVERSITA DEGLI STUDI DI MILANO	IT	999995796
4	BEN	HEK	HELEKOPTER AB	SE	929523841
5	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
6	BEN	ALFA	ALFA LAVAL TECHNOLOGIES AB	SE	983194329
7	BEN	KNEIA	KNEIA SL	ES	936881582
8	BEN	WEB	WEBBER BRENNERTECHNIK GMBH	DE	963762416
9	AP	TWI	TWI LIMITED	UK	999912764
10	AP	ARU	ANGLIA RUSKIN UNIVERSITY HIGHER EDUCATION CORPORATION	UK	991657094

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
11	AP	LUN	UNIVERSITY OF LANCASTER	UK	999840984

12	AP	FORD	FORD MOTOR COMPANY LIMITED	UK	989798089

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LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	ENABLING TECHNOLOGY ASSESSMENT	9 - TWI		1	4	D1.1 – State of the art literature review D1.2 – Ethnographic framework report
WP2	DATA COLLECTION (USE CASES)	10 - ARU		5	26	D2.1 – Individual case study reports D2.2 – Summary report that brings together the lessons from all of the cases combined D2.3 – Detailed case study ethnographic notes
WP3	USE CASE DEVELOPMENT: Skills vs Quantified Products	11 - LUN		5	26	D3.1 – Report showing the best practice technology solutions selected D3.2 – Installed hardware for trial manufacture D3.3 – Case study report demonstrating the impact of implementation
WP4	TRAINING NEEDS ASSESSMENT	2 - MITC		9	34	D4.1 – Training requirements report D4.2 – Hardware and Software training manuals D4.3 – System effectiveness report D4.4 – Future skills training content

WP5	UP-SKILL PLATFORM	5 - IRIS		7	36	D5.1 – Design and requirements specification for data capture and representation D5.2 – Data repository design, implementation and population process D5.3 – Decision Support System design, implementation and preliminary testing

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						validation; on-line testing in the use-case scenarios
WP6	EXPLOITATION, DISSEMINATION & COMMUNICATION	7 - KNEIA		1	36	D6.1 – Up-Skill project website D6.2 – Project Videos 1 D6.3 – Project videos 2 D6.4 – PDEC - Plan for the dissemination, exploitation and communication activities 1 D6.5 – PDEC - Report on the dissemination, exploitation and communication activities 2
WP7	PROJECT MANAGEMENT	1 - MDU		1	36	D7.1 – Data management plan
WP8	Ethics requirements	1 - MDU		1	36	D8.1 – OEI - Requirement No. 1

Work package WP1 – ENABLING TECHNOLOGY ASSESSMENT

Work Package Number	WP1	**Lead Beneficiary**	9. TWI
Work Package Name	ENABLING TECHNOLOGY ASSESSMENT		
Start Month	1	**End Month**	4

Objectives

The goal of this WP is to carry out an expert review of emerging enabling technologies for I5.0 with particular focus on interfaces between them.

Description

Task 1.1 (Design) Review of ethnographic approaches, and method planning with use case development (Lead partner ARU; M1-M3)

The research methods used will be ethnographic involving: shadowing, observing, taking photographs, collecting and reviewing relevant documents (e.g. training manuals related to the technology) and interviewing. The data collection must also be collected across multiple case studies by different researchers. In order to co-ordinate this work and generate comparable data, it will be essential to work together as a team to harmonise the research work. A period of knowledge exchange and training will bring together the over-all approach. Time will also be spent agreeing precisely what sort of data and material will be collected – such as identifying types and sources of uncertainty, the materials used in the course of engineers' and managers' work, the categories of skills used, who engineers and managers interact with and why. The researchers will also have to scope out the roles (the role network) across which tasks associated with the production of what we might term the 'focal artefact' are distributed. In the case of each case study the ethnographer responsible will also need time to familiarise themselves with the various technologies they will be involved with.

Task 1.2: Ethnography workshop review (Lead partner ARU; M4-M4)

As noted above in section 1.1, coordinating the ethnographic data collection will be essential. Alongside prior period of training and methods harmonisation a period of review in which the 'rules of engagement' for the ethnography will be written down and agreed.

Task 1.3: Literature review / technology gap identification (Lead partner TWI; M1-M3)

This task will review the state of the art for Industry 5.0 technologies. A thorough literature review will scope the pertinent details of technologies across industries and across readiness's and applicability of organisations, Potential technology gaps to consider include:
- Users current process capabilities.
- Current skills.
- Complexity of operations required.
- Parts or Bill of Materials (BOM) and/or part counts.
- Cycle times
- Processing environment and likely conditions.

Work package WP2 – DATA COLLECTION (USE CASES)

Work Package Number	WP2	**Lead Beneficiary**	10. ARU
Work Package Name	DATA COLLECTION (USE CASES)		
Start Month	5	**End Month**	26

Objectives

The goal of this WP is to develop case studies within manufacturing/production businesses, and to review skilled roles and identify SWOT for I5.0 within them.

Description
Task 2.1: Case study assignments (Lead partner ARU; M5-M9) Before embarking on the ethnographic aspect of the research, a period of initial case study scoping will be undertaken. The results of this will feed directly into the design of the individual case studies in terms of timing and focus. The specific size and nature of the case studies will impact on how the case study data collection is planned. For example, in the UK we have one large case study organisation, which will explore two distinct technologies (warehouse robots and autonomous vehicles) whereas in Milan we have three distinct case study organisations producing craft outputs. Our case studies also sit in distinct categories – whereas the robot and autonomous guided vehicles may reflect elements of our pre-defined categories of AAA and AAH our Milan craft case studies are more likely to reflect the categories HAA and HHA. The different categories will present different issues to focus on. Gaining prior understanding of the detail of the cases will therefore be essential: 1. What output is being produced by the department / organisation in question and how does this reflect on the natureof the production process (e.g. bespoke, low-volume, batch, mass produced). 2. What markets is the organisation is trying to operate in (and how does this reflect on the overall strategic and businesschoices of the case study organisation. For example, is flexibility and quality traded off against cost and waste. 3. What is the precise material nature of the technology that is the subject of the case (the focal technology) and whathuman skills/level of automation are likely to be encountered? In other words, what inputs/skills are required to make it? 4. What roles, tasks and role relationships exist in relation to the production of the focal technology. Task 2.2: Ethnographic research period (Lead partner ARU; M11-M22) Each academic partner will be responsible for data collection from the case studies in their region with oversight from ARU. ARU will also take the role of ensuring coordination between the data collection by the other partners. Delivery of case-study ethnographies by academic individual partners - Each case study partner will employ one postgraduate researcher to conduct the ethnographic work and then supportthe production of outputs. They will also provide a supervisory team to support and guide the work and - The ethnographic research period will run continuously over the period of one-year. In each region the amount of timespent on site will depend on the number of employees and related roles). - The ethnographies will begin at the case study site after enabling technologies are in place. - The post graduate researcher will be present at their case study research site(s) three days per week. - The specific design of production may occasion visits to other sites as the case study develops – such as in cases wheretightly otherwise coupled-production processes take place on different sites. - Days not spent on site will be used writing up observations notes and interviews Case study coordination and learning between cases ARU will ensure the coordination of the individual case-studies to ensure that a) comparable data is collected and b) cross-case learning is achieved. - Quarterly workshops will be held, face to face or online, in which researchers and their local academic supervisors willpresent progress and insights to the whole team for reflection and discussion. This will shape the ongoing work of the case studies, ensure comparability between cases and feed into final outputs of all of the cases. - ARU will be responsible for recording, storing and integrating workshop insights. - Partners will be responsible for storing ethnographic data. Each use-case ethnography will involve: - Role shadowing, observation, taking fieldnotes in a prescribed and agreed manner (translated into English wherenecessary) and taking photographs where possible. - Interviewing will also be used as a further means to gain insights into how engineers and skilled artisans do their work.Semi-structured qualitative interviews (30-60 minutes) will be used. These will also be the primary means for accessing managers responsible for design and production work. - Interviews will also be the primary means of collecting data on the role networks in which - Documentary analysis of company archives and publicly available documents (local/national newspaper articles,books). Written field notes will also be augmented and improved off-site (e.g. sentences completed, technical terms checked for spelling and augmented with supplementary explanations). This will be done to make the notes as accessible as possible to the rest of the team. Two days per week are set aside for this work. Task 2.3: Detailed case study write-up for Up-Skill platform (Lead partner ARU; M22-M26) - The partner will be responsible for writing up the case studies in their region following an agreed format.

- Each case write-up will identify how roles are lost, transformed and have content transferred elsewhere as aconsequence of technology change.

- ARU responsible for coordinating the inputs of partners in the writing of:
  - the initial state of the alert literature review
  - a summary report which brings together main learning points and best practicesfrom across the cases showing

Work package WP3 – USE CASE DEVELOPMENT: Skills vs Quantified Products

Work Package Number	WP3	**Lead Beneficiary**	11. LUN
Work Package Name	USE CASE DEVELOPMENT: Skills vs Quantified Products		
Start Month	5	**End Month**	26

Objectives

The goal of this WP is to develop case studies within supply chain settings and to review 'hand built' products with respect to opportunities for data-driven improvements in yield/ultimate performance

Description

Task 3.1: Identification of enabling technologies (Lead partner LUN; M5-M7)

The information gained in WP1 will facilitate the provision of a framework to be used for the development of the different use cases. The case study development will follow a 4-step process, the first of which is the identification of technologies and process conditions for each of the case studies to be developed. The overall technology identification process for each case study will require us to first establish the product or sub-assembly to be made as well as the processing steps to be undertaken to develop the selected part. Ethnographic techniques will be used to monitor and assess the material handling process, with particular scrutiny on the route for materials handling and flow paths. This will enable the identification of the technologies and processing conditions to meet the requirements for each of the case studies.

Task 3.2: Manufacturing workflows (Lead partner LUN; M7-M9)

With the identification of the key enabling technologies in T3.1, the next stage will be to detail the granular activitylevel steps that must be completed to create finished goods (products or sub-assemblies identified in T3.1) from the time raw materials are received at the facility until those materials are turned into finished goods for each use case through analysis of the manufacturing workflows. This will require us to identify the likely material requirements as well as the assessment of machine processing conditions and materials in and out puts. We will also analyse the pre and post processing conditions and the finishing and packaging options for the products/sub-assemblies selected for each case study. Finally, we assess the impact of ethnography on the identified conditions and likely system improvements

Task 3.3: Hardware identification and installation (Lead partner TWI; M8-M12)

Task 3.3 will identify the suitable processing and appropriate hardware to meet the requirements for each case study and support its installation. The equipment requirements will be assessed with key stakeholders according to the ethnography findings and then we will shortlist the hardware and confirm pricing, T&Cs and reconfirm findings/agreements. The hardware will then be ordered and installed to align with the WP4 programming.

Task 3.4: Trial manufacture (with ethnography) (Lead partner TWI; M11-M24)

This task will see to the implementation of the case studies based on findings from the previous tasks so as to determine the impacts of ethnography based manufacture. For each case study, we will develop a manufacturing strategy, with the manufacturing aligned to organizational and behavioural characteristic to develop unique processing conditions and possible end-product eccentricities. The manufacturing trials will be run and the resulting products will be tested to obtain feedback from the case study owner, with iterative feedback to the trials. For each case study and base on previous findings.

Task 3.5: Trial feedback back case study (Lead partner LUN; M25-M26)

In this task, the results and feedback from the different case studies will be gathered and collated for review and

reporting with stakeholders. Additional feedback from the stakeholders will also be captured, together with ethnography characteristics and programming options for use in the Up-Skill platform.

Work Package Number	WP5	**Lead Beneficiary**	5. IRIS

Work Package Name	UP-SKILL PLATFORM

Work package WP4 – TRAINING NEEDS ASSESSMENT

Work Package Number	WP4	**Lead Beneficiary**	2. MITC
Work Package Name	TRAINING NEEDS ASSESSMENT		
Start Month	9	**End Month**	34

Objectives

The goal of this WP is extract training needs from previous WP and translate them into training programmes to address broader industry.

Description

Task 4.1: Digital outcomes and training requirements (Lead partner LUN; M9-M14)

Each case study will require its own set of processing conditions, in conjunction with particular ethnography requirements. These processing conditions and requirements will enable the bespoke alignment of programming and training to address broader industry. We are going to create customised programming for each part or subassembly process for each case study's respective manufacturing cells.

Task 4.2: Programme training and usage methods (Lead partner MITC; M12-M25)

For each case study, there will be need for training to use the software packages, for manipulation of the software to accommodate changes and for production of different parts. This task will see to the generation of the training manual according to each software to be used. The task will also see to the conducting of a relevant course for the partners and the overseeing of the in-service conditions and usage by case studies.

Task 4.3: Hardware training (Lead partner TWI; M14-M25)

For each case study, there will be need for training to use hardware, for manipulation of the hardware to accommodate changes and for production of different parts. In this task, we will generate a training manual for each of the hardware (equipment) to be used for the different case studies. The task will also see to the conducting of a relevant course for the partners and the overseeing of the in-service conditions and usage by case studies.

Task 4.4: Evaluation of system effectiveness (Lead partner MITC; M26-M30)

This task will review the implemented software and hardware changes across case studies. To qualify the changes are robust, and effective a FATs (Factory Acceptance Testing) methodology will be developed. The FATs document will form a checklist of activities to execute across all use cases with stakeholders in order to pass off implemented changes and fully evaluate the system

Task 4.5: Future skills mapping (Lead partner UMIL; M27-M34)

Each of the case studies will be in a position to assess how existing skilled roles are being, and might further, augmented by technology. We will identify the new skills that existing craft workers will need to take on in order to survive in the emerging digitalised workplace. Thus, unlike typical skills-mapping exercises, our skills-mapping will focus on 'role survival' and 'role re-distribution', rather than the identifying the skills required for newly emerging roles (such as, for example, coding).

Work package WP5 – UP-SKILL PLATFORM

Start Month	7	**End Month**	36

Objectives

Design and requirements definition for data capture and representation. Develop data repository, data capture processes and pipelines, populate with data. Develop Decision Support System (DSS) using advanced data processing algorithms (AI techniques) Validation testing offline/in-laboratory with KPIs. Validation testing on-line/in-situ with KPIs and integration with use cases.
Description
Task 5.1 Design and requirements definition for data capture and representation (Lead partner IRIS; M7-M12) A study will be performed of all the data sources and requirements for the project. One the one hand, what data sources and prior reports, studies and existing repositories will be considered to study; and will be necessary for the use cases to capture human and robot/machine relationships in the different environments. On the other hand, knowledge in different formats and levels (Political, educational, economic, environmental, etc.…) which describe the data sources for specific tasks and skills needed by the end users; and considering the evolution/creation/destruction of jobs. The design will also make use of the information provided from the WP1 assessment in order to produce a compendium of micro-architectures supporting a global backend-frontend system, including a top user-friendly layer for practical and satisfying experience. Task 5.2 Data repository development, data capture processes and pipelines, data population and formatting (Lead partner IRIS; M10-M24) Using the specifications developed in T5.1, the data repository and data capture processes will be implemented. A Case Based Reasoning system (based on Artificial Intelligence principles) will provide retrieval of the data, using appropriate distance functions to find historical cases closest to new scenarios. Also, soft-sensors can be developed to provide enhanced information by processing and aggregating the raw data. This will be performed in collaboration and synchronization with WP3 use case development, also using SWOT methodology for each use case. Task 5.3 Decision Support System (DSS) development using advanced data processing algorithms (AI techniques) (Lead partner IRIS; M14-M36) T5.3.1: DSS and AI algorithms development: Employing the state-of-the-art numerical techniques and efficient algorithms, raw data will be processed into higher level information, and fused with different data sources, related to the KPIs. Key KPIs will include productivity (time, cost, safety, waste reduction), worker satisfaction, risk assessment, among others. Weighted aggregation operators (based on Artificial Intelligence principles) will be used to process data from different existing sources, taking into account reliability and relevance to each of the KPIs and end-user needs. Data algorithms will include those KPI considered critical for the industrial jobs' transformation in the context of the fourth industrial revolution (with latency of 30 frames per second). This will be performed in collaboration and synchronization with WP3 use case development. T5.3.2: Validation of digital tools under lab conditions and in association with AI developed techniques: The systems previously developed will be validated offline with available real historical data which can be supplemented with simulated data as necessary. As the project progresses more data will become available and so this task will have an iterative nature, involving all participant partners. Task 5.4 Validation testing on-line/in-situ with KPIs and integration with use cases (Lead partner IRIS; M28-M36) The systems developed in Tasks 5.2 and 5.3 will be validated online in the real use-case environments. The system will be deployed and integrated into the use case environment with installed computer servers via the digital platform for local data processing. This will again be performed in collaboration and synchronization with WP3 use case development.

Work Package Number	WP6	**Lead Beneficiary**	7. KNEIA

Work Package Name	EXPLOITATION, DISSEMINATION & COMMUNICATION		
Start Month	1	**End Month**	36

Work package WP6 – EXPLOITATION, DISSEMINATION & COMMUNICATION

Objectives
The overall WP objectives are: Enable potential future exploitation of the project results by disseminating them to the relevant stakeholders. Ensure project's findings are widely communicated to the scientific community and the general public. Document undertaken and proposed dissemination and communication activities. Ensure the project results reach the relevant stakeholders who will use and implement them. Guarantee IPR management according to the Guidelines of the EC within the Consortium Agreement Ensure clustering and coordination with other related EU/National projects and initiatives. Attract business partners and potential stakeholders and to create market demand.
Description

Task 6.1 Communication Activities (Lead partner KNEIA; M1-M36)

T6.1.1. Communication Plan and reporting on communication: KNEIA will draw up a Communication Plan, as part of the PDEC (D6.3), to mobilise stakeholders' interests and secure their support to the project. This plan will be closely linked with the dissemination activities and it will follow the steps described in section 2.2.
T6.1.2 Project website: The website (D6.1) will serve as communication tool to all target audiences and will house a restricted access area for partners to store information and facilitate file sharing and internal communication. The public site will present the brand and non-confidential information: concept, partners, core objectives, infographics on the strategic, competitive and economic benefits, etc. The platform will enable to disseminate non-confidential results (reports, deliverables, publications as part of our green open access, etc.). At the end of the project, public information and results will be kept as the central virtual hub for post-project activities.
T6.1.3 Communication and promotional materials: Project leaflets, posters, project video (D6.2), newsletters and promotional materials (gadgets for tradeshows, etc.) will be designed to support communication and branding efforts. At least 2 dissemination videos – an introductory video on month 6 for the general public and a divulgative report video at the end of the project describing the final results- will be developed and published on YouTube, the project website, and social media, to raise awareness and engage stakeholders with project pilots and to promote uptake after the end of the project. These materials will be annexed to the PDEC (D6.3).
T6.1.4 Social/digital media channels: A social media grid will be drawn up and will plot the most appropriate channels for targeting specific audiences. Pages will be designed for each channel and a content maintenance plan will be put in place for managing the stream of information to secure and maintain followers. Social and digital media will contribute to create 'communities of support' to the project. Tools such as Twitter, LinkedIn and YouTube will be used. Visual media, videos, animations, icons, info-graphics and mobile enabled content and richer content experiences will be sought.
Task 6.2 Dissemination Activities (Lead partner KNEIA; M1-M36)
A dissemination plan will be developed as part of the PDEC, building on the preliminary plan in Section 2.2 and in complete alignment with the principles of Open Science. The following activities will be executed:

- Publishing at least 6/8 scientific articles in journals and conferences (after preliminary IP protection check);
- Organising special sessions and scientific workshops as part of large conferences with high impact;
- Targeted placement of articles and features in relevant trade and industry focused publications;
- Organisation of the "Up-Skill Agenda" of events spanning the project duration: trade fairs, exhibitions, workshops,seminars, conferences, round tables, etc., that the partners are already planning to attend/organise within ongoing activities. The target is 24 existing events and organisation of: at least 2 intermediate workshops with brother projects and a Final conference (M36), to showcase results and impacts.

Task 6.3 Exploitation of results, IPR and Business model (Lead partner KNEIA; M12-M36)

A draft of the Exploitation Plan will be delivered as part of the PDEC (D6.3). Built on the information already provided in Section 2.2, the Exploitation Plan will be structured as follows:

- The planned key exploitable results and their expected key applications: development status, facts and figures thatfacilitate the assessment of the potential impact, differences from existing competing products/services, main IPR issues to be clarified in the consortium agreement;
- The exploitation team: partners involved in the exploitation activities providing information on their previousexperience and business interests, complementing external resources if not in the consortium;
- The market: the potential geographical and economic size of the market, market trends, main competitors andcompetitive advantage;
- Business model including marketing strategy and main clients: will be elaborated using CANVAS methodologyadapted to the relevant target markets.

Task 6.4 Clustering and Networking activities (Lead partner KNEIA; M1-M36)

This task will ensure that Up-Skill is networked with relevant national and international projects, initiatives and networks. Networking with other projects will ensure knowledge interchange and reuse between previous and future projects. A close interaction with other proposals under the same or closer topics will be ensured. The
networking plan will take into account projects within international stakeholders and other EU/national projects (as identified in Section 1.2). Most of the project partners are members of relevant networks, groups and associations. These memberships will promote cooperation and integration with Up-Skill. Members of consortia of the networked projects will be invited to the 2 intermediate workshops (T6.2) and exchange of (non-sensitive) information will be implemented (e.g., by newsletter or dedicated meetings). At least 2 dissemination events in cooperation with other related projects will be organised and included in the Up-Skill Agenda. One section of the

Newsletter (T6.1) will be dedicated to the information on other ongoing projects and exchange of newsletters from the other connected projects will be pursued. The progress of this task will be reported in the PDEC.

Task 6.5 Stakeholder analysis and engagement (Lead partner KNEIA; M1-M36)

The stakeholder analysis (stakeholder mapping) will identify the most important stakeholders and end-users of the UpSkill technologies and assess their position towards the project's results in order to set up engagement strategies. In addition, the partners will jointly work to identify and engage further stakeholders and end-users
for Up-Skill technological solutions throughout the duration of the project. All stakeholders identified in the initial stakeholder analysis will be invited to participate in an online survey designed to measure stakeholder characteristics, such as their interest, attitude, influence, knowledge and expectations relevant for the project.
Based on the outcome of the survey, more targeted dissemination and exploitation actions will be implemented and relationships built with key stakeholders to the project. Relevant feedback about stakeholders' expectations will be constantly provided to the project technological developers to be taken into account in the technology definition and upgrade. Survey results will inform the annual revision of the PDEC.

Work package WP7 – PROJECT MANAGEMENT

Work Package Number	WP7	**Lead Beneficiary**	1. MDU
Work Package Name	PROJECT MANAGEMENT		
Start Month	1	**End Month**	36

Objectives

This work package deals with the coordination and management of all project activities and reporting to the EC.
The project will be coordinated by MDU, with the help of a Management Committee, with a representative from each consortium partner, which will lead the management of the project, having the authority to make decisions on project execution.
- To coordinate the technological and scientific orientation of the project, to ensure the delivery of the project on timeand within the budget and that the project maintains its relevance towards the objectives defined in Section 1.1 and within the context of each WP.
. To manage resources, monitor the overall project performance and manage risks (identified in Table 3.2b and any that may arise during the project), ethics, and contingencies.
- Control cost expenditure within authorised budgets for each period by Task/WP and for the entire project

Description

Task 7.1 Technical Management (Lead partner TWI, all partners; M1-M36)

The technical management task will be responsible for monitoring the progress of the technical outcomes of the project, with emphasis given on project milestones and deliverables. Steven Baines (Technical Manager) will regularly monitor progression of work activities, ensuring that both the general project objectives defined in Section 1.1 and the individual work package objectives are met by comparing the estimated baseline against actual performance. Table 3.2a of milestones will be used to check the broad status of the project using the defined means of verification. The Gantt, will be useful for an in-depth assessment of project progress (to WP and task-levels) with the deliverables given in Table 3.1c being a key indicator of project status. Using the Gantt, tasks- and sub-task descriptions in WPs and deliverables,

the actual status of the project at any specific time can be compared to baseline schedule to determine if activities are ahead or behind schedule. This will allow for reprioritisation of remaining work (if need be) or management of changes to project baseline if necessary.
Within the task, a project quality board will be defined that will undertake the quality assurance of the project outcomes. More-over, by continuously monitoring the progress of the project and its alignment with the initial objectives, this task will perform also risk management activities, in order to apply contingency actions when necessary. Regarding the risk management, the coordinator of the project will: i) ensure the quality of the developed work throughout the project. For that purpose, this task will set and implement a quality process covering the quality assurance and control aspects. This will include the definition of templates for documentation and presentations as well as policies for external communication by individual partners on
project matters, ii) set appropriate mechanisms to identify, anticipate, communicate and mitigate potential risks and deviations in the project, iii) ensure the constant and effective workflow among all project participants, formulating and implementing a project framework and create related tools for use in the entire project. This
task includes the set-up of suitable collaboration and knowledge management and sharing tools to be used within the project.

Task 7.2 Financial Management (Lead partner MDU, all partners; M1-M36)

The Project Coordinator will be responsible for the financial coordination of the project. This task includes establishing and maintaining financial records, coordination of costs submission, preliminary checks of individual costs against known criteria (contractual commitments, progress reports, and delivery of results) and
consolidation of costs, follow-up of EC payments, and delivery of all financial documents required periodically by the EC. The PC will also follow up the project expenses and track possible deviations. The PC will also be responsible for distributing the EC contribution amongst the partners according to the Grant Agreement and the Consortium Agreement.

Task 7.3 Project Reporting (Lead partner MDU, all partners; M1-M36)

This task deals with the liaison with the EC through the set reporting mechanisms (bi-annual management reports to EC, annual Periodic review reports-APRR, final report). The Periodic Management Reports summarize organizational documentation and planning progress on a management level, which is not covered by the documentation in the individual working groups. The project final report is a detailed summary of the periodic progress reports including management as well as scientific aspects, the main results of the project and a general evaluation of the project outcome from the perspective of the consortium.

Work package WP8 – Ethics requirements

Work Package Number	WP8	**Lead Beneficiary**	1. MDU
Work Package Name	Ethics requirements		
Start Month	1	**End Month**	36
Objectives			
The objective is to ensure compliance with the 'ethics requirements' set out in this work package.			
Description			
This work package sets out the 'ethics requirements' that the project must comply with.			

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (automatically posted online)
Sensitive — limited under the conditions of the Grant Agreement
EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	State of the art literature review	WP1	11 - LUN	R — Document, report	SEN - Sensitive	3
D1.2	Ethnographic framework report	WP1	10 - ARU	R — Document, report	SEN - Sensitive	4
D2.1	Individual case study reports	WP2	10 - ARU	R — Document, report	SEN - Sensitive	9
D2.2	Summary report that brings together the lessons from all of the cases combined	WP2	10 - ARU	R — Document, report	SEN - Sensitive	22
D2.3	Detailed case study ethnographic notes	WP2	10 - ARU	R — Document, report	SEN - Sensitive	26
D3.1	Report showing the best practice technology solutions selected	WP3	11 - LUN	R — Document, report	SEN - Sensitive	7
D3.2	Installed hardware for trial manufacture	WP3	9 - TWI	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	12
D3.3	Case study report demonstrating the impact of implementation	WP3	11 - LUN	R — Document, report	SEN - Sensitive	26

D4.1	Training requirements report	WP4	2 - MITC	R — Document, report	PU - Public	14
D4.2	Hardware and Software training manuals	WP4	11 - LUN	R — Document, report	SEN - Sensitive	25
D4.3	System effectiveness report	WP4	2 - MITC	R — Document, report	SEN - Sensitive	30
D4.4	Future skills training content	WP4	3 - UMIL	R — Document, report	PU - Public	34

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.1	Design and requirements specification for data capture and representation	WP5	5 - IRIS	R — Document, report	SEN - Sensitive	12
D5.2	Data repository design, implementation and population process	WP5	5 - IRIS	R — Document, report	SEN - Sensitive	24
D5.3	Decision Support System design, implementation and preliminary testing validation; on-line testing in the use-case scenarios	WP5	5 - IRIS	R — Document, report	SEN - Sensitive	36
D6.1	Up-Skill project website	WP6	7 - KNEIA	DEC —Websites, patent filings, videos, etc	PU - Public	6

D6.2	Project Videos 1	WP6	7 - KNEIA	DEC —Websites, patent filings, videos, etc	PU - Public	6
D6.3	Project videos 2	WP6	7 - KNEIA	DEC —Websites, patent filings, videos, etc	PU - Public	36
D6.4	PDEC - Plan for the dissemination, exploitation and communication activities 1	WP6	7 - KNEIA	R — Document, report	PU - Public	6
D6.5	PDEC - Report on the dissemination, exploitation and communication activities 2	WP6	7 - KNEIA	R — Document, report	PU - Public	36
D7.1	Data management plan	WP7	1 - MDU	DMP — Data Management Plan	PU - Public	3

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D8.1	OEI - Requirement No. 1	WP8	1 - MDU	ETHICS	SEN - Sensitive	6

Description

The Deliverable will cover
1. What output is being produced by the department / organisation in question and how does this reflect on the natureof the production process (e.g. bespoke, low-volume, batch, mass produced).
2. What markets is the organisation is trying to operate in (and how does this reflect on the overall strategic and businesschoices of the case study organisation. For example, is flexibility and quality traded off against cost and waste.
3. What is the precise material nature of the technology that is the subject of the case (the focal technology) and whathuman skills/level of automation are likely to be encountered? In other words, what inputs/skills are required to make it?
4. What roles, tasks and role relationships exist in relation to the production of the focal technology.

Deliverable D1.1 – State of the art literature review

Deliverable Number	D1.1	**Lead Beneficiary**	11. LUN
Deliverable Name	State of the art literature review		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP1
Description			
This Deliverable will describe the state of the art for Industry 5.0 technologies. A thorough literature review will scope the pertinent details of technologies across industries and across readiness's and applicability of organisations, Potential technology gaps to consider include: Users current process capabilities. Current skills. Complexity of operations required. Parts or Bill of Materials (BOM) and/or part counts. Cycle times Processing environment and likely conditions.			

Deliverable D1.2 – Ethnographic framework report

Deliverable Number	D1.2	**Lead Beneficiary**	10. ARU
Deliverable Name	Ethnographic framework report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	4	**Work Package No**	WP1
Description			
The research methods used will be ethnographic involving: shadowing, observing, taking photographs, collecting and reviewing relevant documents (e.g. training manuals related to the technology) and interviewing. The data collection must also be collected across multiple case studies by different researchers. The 'rules of engagement' for the ethnography will be written down and agreed.			

Deliverable D2.1 – Individual case study reports

Deliverable Number	D2.1	**Lead Beneficiary**	10. ARU
Deliverable Name	Individual case study reports		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	9	**Work Package No**	WP2

Description
Report showing the best practice technology solutions selected

Deliverable D3.2 – Installed hardware for trial manufacture

Deliverable Number	D3.2	**Lead Beneficiary**	9. TWI
Deliverable Name	Installed hardware for trial manufacture		

Deliverable D2.2 – Summary report that brings together the lessons from all of the cases combined

Deliverable Number	D2.2	**Lead Beneficiary**	10. ARU
Deliverable Name	Summary report that brings together the lessons from all of the cases combined		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	22	**Work Package No**	WP2
Description			
Detailed case study write-up for Up-Skill platform Writing up of the case studies in the separate regions. Each case write-up will identify how roles are lost, transformed and have content transferred elsewhere as a consequence of technology change. Inputs of partners in the writing of: o the initial state of the alert literature review o a summary report which brings together main learning points and best practices from across the cases showing			

Deliverable D2.3 – Detailed case study ethnographic notes

Deliverable Number	D2.3	**Lead Beneficiary**	10. ARU
Deliverable Name	Detailed case study ethnographic notes		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	26	**Work Package No**	WP2
Description			
Each academic partner will be responsible for data collection from the case studies in their region. Delivery of casestudy ethnographies by academic individual partners a) comparable data should be collected and b) cross-case learning should be achieved.			

Deliverable D3.1 – Report showing the best practice technology solutions selected

Deliverable Number	D3.1	**Lead Beneficiary**	11. LUN
Deliverable Name	Report showing the best practice technology solutions selected		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	7	**Work Package No**	WP3
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP3
Description			

Installed hardware for trial manufacture The hardware will be ordered and installed to align with the WP4 programming.

Deliverable D3.3 – Case study report demonstrating the impact of implementation

Deliverable Number	D3.3	**Lead Beneficiary**	11. LUN
Deliverable Name	Case study report demonstrating the impact of implementation		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	26	**Work Package No**	WP3
Description			
Case study report demonstrating the impact of implementation			

Deliverable D4.1 – Training requirements report

Deliverable Number	D4.1	**Lead Beneficiary**	2. MITC
Deliverable Name	Training requirements report		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	14	**Work Package No**	WP4
Description			
Training requirements report Digital outcomes and training requirements			

Deliverable D4.2 – Hardware and Software training manuals

Deliverable Number	D4.2	**Lead Beneficiary**	11. LUN
Deliverable Name	Hardware and Software training manuals		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	25	**Work Package No**	WP4

Deliverable D4.3 – System effectiveness report

Deliverable Number	D4.3	**Lead Beneficiary**	2. MITC
Deliverable Name	System effectiveness report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive

Description
Programme training and usage methods For each case study, there will be need for training to use the software packages, for manipulation of the software to accommodate changes and for production of different parts. For each case study, there will also be need for training to use hardware, for manipulation of the hardware to accommodate changes and for production of different parts.

Due Date (month)	30	**Work Package No**	WP4
Description			
System effectiveness report This Deliverable will review the implemented software and hardware changes across case studies. To qualify the changes are robust, and effective a FATs (Factory Acceptance Testing) methodology will be developed. The FATs document will form a checklist of activities to execute across all use cases with stakeholders in order to pass off implemented changes and fully evaluate the system			

Description

Design and requirements specification for data capture and representation THe Deliverable will cover all the data sources and requirements for the project. One the one hand, what data sources and prior reports, studies and existing repositories will be considered to study; and will be necessary for the use cases to capture human and robot/machine relationships in the different environments. On the other hand, knowledge in different formats and levels (Political, educational, economic, environmental, etc.…) which describe the data sources for specific tasks and skills needed by the end users; and considering the evolution/creation/destruction of jobs. The design will also make use of the information provided from the WP1 assessment in order to produce a compendium of micro-architectures supporting a global backend-frontend system, including a top user-friendly layer for practical and satisfying experience.

Deliverable D4.4 – Future skills training content

Deliverable Number	D4.4	**Lead Beneficiary**	3. UMIL
Deliverable Name	Future skills training content		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	34	**Work Package No**	WP4
Description			
Future skills training content Each of the case studies will be in a position to assess how existing skilled roles are being, and might further, augmented by technology. We will identify the new skills that existing craft workers will need to take on in order to survive in the emerging digitalised workplace. Thus, unlike typical skills-mapping exercises, our skills-mapping will focus on 'role survival' and 'role re-distribution', rather than the identifying the skills required for newly emerging roles (such as, for example, coding).			

Deliverable D5.1 – Design and requirements specification for data capture and representation

Deliverable Number	D5.1	**Lead Beneficiary**	5. IRIS
Deliverable Name	Design and requirements specification for data capture and representation		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP5

Deliverable D5.2 – Data repository design, implementation and population process

Deliverable Number	D5.2	**Lead Beneficiary**	5. IRIS
Deliverable Name	Data repository design, implementation and population process		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP5
Description			

Task 5.2 Data repository development, data capture processes and pipelines, data population and formatting (Lead partner IRIS; M10-M24) Using the specifications developed in T5.1, the data repository and data capture processes will be implemented. A Case Based Reasoning system (based on Artificial Intelligence principles) will provide retrieval of the data, using appropriate distance functions to find historical cases closest to new scenarios. Also, soft-sensors can be developed to provide enhanced information by processing and aggregating the raw data. This will be performed in collaboration and synchronization with WP3 use case development, also using SWOT methodology for each use case.

Deliverable Number	D6.1	**Lead Beneficiary**	7. KNEIA
Deliverable Name	Up-Skill project website		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP6
Description			

Up-Skill project website
The website (D6.1) will serve as communication tool to all target audiences and will house a restricted access area for partners to store information and facilitate file sharing and internal communication.
The public site will present the brand and non-confidential information: concept, partners, core objectives, infographics

Deliverable D5.3 – Decision Support System design, implementation and preliminary testing validation; on-line testing in the use-case scenarios

Deliverable Number	D5.3	**Lead Beneficiary**	5. IRIS
Deliverable Name	Decision Support System design, implementation and preliminary testing validation; online testing in the use-case scenarios		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5
Description			
Decision Support System design, implementation and preliminary testing validation; on-line testing in the use-case scenarios The deliverable includes A description of the DSS and AI algorithms development Validation of digital tools under lab conditions and in association with AI developed techniques Validation testing on-line/in-situ with KPIs and integration with use cases			

Deliverable D6.1 – Up-Skill project website

on the strategic, competitive and economic benefits, etc. The platform will enable to disseminate non-confidential results (reports, deliverables, publications as part of our green open access, etc.). At the end of the project, public information and results will be kept as the central virtual hub for post-project activities.

Deliverable D6.2 – Project Videos 1

Deliverable Number	D6.2	**Lead Beneficiary**	7. KNEIA
Deliverable Name	Project Videos 1		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP6
Description			
A project video will be launched and published			

Description

PDEC - Plan for the dissemination, exploitation and communication activities

The dissemination and communication plan will cover
- Publishing at least 6/8 scientific articles in journals and conferences (after preliminary IP protection check);
- Organising special sessions and scientific workshops as part of large conferences with high impact;
- Targeted placement of articles and features in relevant trade and industry focused publications;
- Organisation of the "Up-Skill Agenda" of events spanning the project duration: trade fairs, exhibitions, workshops,seminars, conferences, round tables, etc., that the partners are already planning to attend/organise within ongoing activities. The target is 24 existing events and organisation of: at least 2 intermediate workshops with brother projects and a Final conference (M36), to showcase results and impacts.
- Also an networking and stakeholder plan will be included.

Deliverable D6.3 – Project videos 2

Deliverable Number	D6.3	**Lead Beneficiary**	7. KNEIA
Deliverable Name	Project videos 2		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
A second project video will be launched and published			

Deliverable D6.4 – PDEC - Plan for the dissemination, exploitation and communication activities 1

Deliverable Number	D6.4	**Lead Beneficiary**	7. KNEIA
Deliverable Name	PDEC - Plan for the dissemination, exploitation and communication activities 1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP6
The Exploitation plan will cover: The planned key exploitable results and their expected key applications: development status, facts and figures that facilitate the assessment of the potential impact, differences from existing competing products/services, main IPR issues to be clarified in the consortium agreement; The exploitation team: partners involved in the exploitation activities providing information on their previous experience and business interests, complementing external resources if not in the consortium; The market: the potential geographical and economic size of the market, market trends, main competitors and competitive advantage; Business model including marketing strategy and main clients: will be elaborated using CANVAS methodology adapted to the relevant target markets.			

Deliverable D6.5 – PDEC - Report on the dissemination, exploitation and communication activities 2

Deliverable Number	D6.5	**Lead Beneficiary**	7. KNEIA
Deliverable Name	PDEC - Report on the dissemination, exploitation and communication activities 2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			

PDEC - Report on the dissemination, exploitation and communication activities

The dissemination and communication report will cover all performed activities in the dissemination, exploitation and communication plan.

Deliverable D7.1 – Data management plan

Deliverable Number	D7.1	**Lead Beneficiary**	1. MDU
Deliverable Name	Data management plan		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	3	**Work Package No**	WP7
Description			

The DMP will cover:

- Types of data/research outputs: definition of dataset units to be produced during the project life span. Each data setwill contain, at least the work package to which it belongs and data typology, with a short description.
- Findability of data/research outputs: datasets will be stored on Zenodo, automatically becoming part of OpenAIRE,the EC-funded initiative supporting the Open Access policy. Each will be assigned a unique identifier.
- Accessibility of data/research outputs: use restrictions and licensing information. Open Access will be implementedin peer-review publications, conference proceedings and workshop presentations.
- Interoperability of data/research outputs: standards used to describe datasets. Proper metadata facilitates the use ofthe data by others, makes it easier to combine information from different sources and ensures transparency. Metadata of deposited data will be open under a Creative Common Public Domain Dedication (CC 0).
- Reusability of data/research outputs: the right to access and reuse digital research data under the terms set out in theGrant Agreement will be ensured. The DMP will include information on licensing of data, their availability, re-use of data, duration of data for re-use and reproducibility of research outputs, among other issues.
- Curation and storage/preservation: The Exploitation manager will oversee data management, while quality control ofeach dataset is the responsibility of every WP leader. Informed consent in data collection and information protection in data storage and access will be considered, as well as procedures for longterm preservation of datasets.

Deliverable D8.1 – OEI - Requirement No. 1

Deliverable Number	D8.1	**Lead Beneficiary**	1. MDU
Deliverable Name	OEI - Requirement No. 1		
Type	ETHICS	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP8
Description			

An involvement of an external and independent ethical advisor, who has adequate familiarity with ethnographic research and ethical issues, is strongly called for. The ethics advisor will have to identify all ethical issues related to the project, give guidance to achieve adequate handling, and report the activities and compliance with EU and national regulations.

The proposal states that a data management plan (DMP) will be produced within the first six months of the project. This will enable the collection of data generated throughout this project and will be essential for achieving any regulatory approval. However it is not sufficiently detailed in the Ethics Self-assessment nor in the proposal, a full description of ethics procedures for the following ethics issues:

Section 4: PERSONAL DATA
Development of case study requirements within manufacturing/production businesses, reviewing skilled roles, customs, habits and cultural difference in their approach and identifying SWOT for I5.0 within them. The generated data (per each case) feeds in to WP3 and the Up-Skill platform.

Section 6: NON-EU COUNTRIES
Potential risks: privacy issues, performance measurement.

Section 8: ARTIFICIAL INTELLIGENCE
Up-Skill may imply the use of AI (e.g recommendation systems, predictive algorithms).

Section 9: OTHER ETHICS ISSUES
The proposed research is likely to yield potentially social or behavioural unexpected or incidental findings that may require interventions to safeguard the well being of research participants.

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Opportunities/challenges for implementation identified	WP1	1-MDU	Completion of the gap analysis through the successful analysis of the emerging enabling technologies for I5.0: D1.1 & D1.2	4
2	Development of case studies within manufacturing/production achieved	WP2	1-MDU	Report, identifying opportunities for implementation of I5.0 within the case study scenarios D2.1 & D2.2	26
3	Development of case studies within supply chain settings achieved	WP3	1-MDU	Report, identifying opportunities for implementation of I5.0 within the case study scenarios D3.1 & D3.3	26
4	Training programme content ready to deliver on a broader EU scale	WP4	2-MITC	Hardware and software training modules generated D4.1 & D4.2 & D4.4	30
5	Online Up-Skill platform is activated	WP5	5-IRIS	On-line testing of Up-Skill platform in the usecase scenarios (D5.3)	32
6	Public Awareness & Dissemination achieved	WP6	7-KNEIA	Detailed report on dissemination outcomes D6.4	36
7	Final Exploitation Strategy & Stakeholder Engagement available	WP6	7-KNEIA	Detailed report on the exploitation strategy completed & stakeholder engagement available D6.5	36

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Failure to deliver according to the work plan by either of the partners	WP5, WP8, WP6, WP4, WP1, WP7, WP2, WP3	The has been carefully distributed according to the strengths and expertise of each of the partners to ensure high quality work. Additionally, our project monitoring process includes quarterly meetings that will facilitate prompt identification of delays or failures, allowing the consortium to find additional expertise and assign additional effort when necessary.
2	Partner leaves the consortium	WP6	In case of Partner's resignation pending work-load will be either re-distributed among Partners on the basis of their competencies or a replacement Partner will be identified. In addition, the Consortium Agreement will foresee such situations and will describe measures to be taken to prevent non-compliance to project activities
3	Low quality of project results	WP6	The internal review process for all project deliverables and reports will ensure high quality project results.
4	Delays in the tasks' completion due to lack of resources, inability of partners or disharmony in the collaboration	WP6	The work plan of the project has been carefully planned, and the partners have a lot of experience in participating in such projects. Coordinator and project management team will adopt a proactive approach by identifying all critical paths in the project and by performing the necessary rescheduling and crashing
5	Changes in legislative framework in one or more European Countries	WP6	Continuous monitoring and update of possible adoption of new laws and rules by the European Nations
6	Stakeholders outside the project are not interested	WP6	Stakeholders will be contacted early in the project through various communication activities in an effort to raise interest throughout the scientific and end user community
7	Dissemination not sufficient to create a substantial impact on the targeted stakeholder groups and beyond	WP6	A significant amount of resources is allocated in the work plan to dissemination and exploitation planning (WP5) and thereby guaranteeing the maximum benefit for the partners. The partners already have good access to core stakeholders and channels within academic and commercial circles. We have planned to participate in most major

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			conferences and national/international events as well as organizing our own events to present the project outcomes.
8	Networking activities, Open access to generated data, models and tests jeopardize IP strategy	WP6	TWI will consider and evaluate IP risk before the information is being made public or delivered to stakeholders through the PUDF.
9	Poor quality output use case data to translate into the Up-Skill training programmes	WP1, WP2	Pilot sites have already been carefully selected to ensure that they are suitable for the use cases. T2.1 will analyse the sites and existing infrastructure so as to guarantee the Use Case Requirements during the use case execution. In addition, regular remote meetings will be held to check that all use case teams are aligned
10	Lack of data and/or commitment from dataproducing partners	WP1, WP2	Continuous communication with partners and coordination in order to avoid delays and/or absence of data available.
11	Delays in the platform and/or algorithm development.	WP5	Continuous communication and reporting to the coordination, as well as continuous follow up of schedule and validations.
12	Validation in use cases not successful.	WP3	Escalate the issue to the project owner. Optimization and reshaping of algorithms and digital tool.

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	Timing (month)	Location	Comments
RV1	19	tbd	
RV2	36	tbd	

History of Changes

Part A

Version date	Changes
2022-04-11	Adding researchers (completion of information related with career stage, role and ORCID personal identification number)
	Modification of MDU financial information(raise in the personal cost from 310,764.00 € to 377,021.00 € which automatically led to a raise of the indirect costs from 86,521.00 € to 103,085.00 € ; and therefore change in the "requested EU contribution" to to the maximum amount 515,426.25 €
	Modification of TWI financial information with the corresponding reduction.
	Adding of workpackages and efforts accordingly to the proposal
	Adding of related deliverables
2022-04.12	Modification of the effort in the WP7 regarding MDU and TWI ; instead of 2 (proposal) MDU got 12 and TWI instead of 16 (proposal) got 6. The total effort of the WP doesn't change it's still 39 (like in the proposal)
	Adding of deliverables in the WP6 because two deliverables had two due dates so separated them into two deliverables to keep the two due dates. The deliverables that are concerned were : → D6.2 "project videos"due M6,M36 that now became D6.2 "project videos 1" due M6 and D6.3 "project videos 2" due M36 → D6.3 "PDEC - Plan for the dissemination, exploitation and communication activities" due M6,M36 that now became D6.4 "PDEC - Plan for the dissemination, exploitation and communication activities 1" due M6 and D6.5 "PDEC - Plan for the dissemination, exploitation and communication activities 2" due M36
	Deleting of one deliverable the D7.2 "final project report" due M36
	For deliverable D4.3 "System effectiveness report" the level of dissemination CO wasn´t found on Funding and tenders portals so it was changed to SEN instead
	Adding of all the milestones according to the proposal but the lead beneficiary wasn't on the proposal so for each one of them the lead Beneficiary is the same as of the related WP
	Adding of the critical risks accordingly to the proposal
2022-04-13	Adding of deliverables that could serve as justification for passed milestones
2022-05-05	Breakdown of the efforts in the WP1, WP2, WP3, WP4, WP5, WP6, and WP7 to make them correspond to the sum of partners work according to the Annex 1 description of action (part A) draft
	Adding of the lead beneficiaries in WP1, WP2 and WP3 according to Annex 1 description of action (part A) draft
	Adding of participating partners in WP8
	Adding of the lead beneficiaries for the deliverables D1, D2, D3, D4, D5, D6, D7, D8, D10 according to the Annex 1 description of action (part A) draft
	Adding of the lead beneficiaries for the Milestone M1, M2 and M3. According to the Annex 1 description of action (part A) draft, the lead beneficiaries were supposed to be respectively TWI, ARU and LUN but on the Funding and tenders portal it was not possible to select these partners, therefore MDU was put as lead beneficiary instead.
	The UK budget was restored and the budgets and efforts of the remaining beneficiaries were rearranged, to allow a larger budget and more PM for the new coordinator

	The UK beneficiaries were changed into Associate Partners and the numbering of participants was accordingly changed
2022-05-13	Changes in the budget and PM of KNEIA and related changes in the budget of MDU
2022-05-17	Allocation of Ztift budget and PM to Helekopter AB as well as transferring en block of the tasks and the resources of Ztift to HEK without any changes. Removal of Ztift from the project WP 5 total effort changed from 81.6 to 80.6.
	WP 6 total effort changed from 83.5 to 72.5. Associated with document Ref. Ares(2022)4648166 - 24/06/20 changed from 39 to 45.
2022-05-25	Modification of the efforts in the WPs back to the proposal version with 2 modifications in WP7 where TWI's 16 PMs go to MDU, and 2 PMs from ARU go to TWI. Total efforts per respective WP is still in accordance to the proposal. Modification of the estimated budget back to the proposal version with 3 modifications: MDUs personnel costs and corecponding indirect costs are increased with TWIs budget for the WP7; ZTIFT's budget transferred to HEK; HEK transferred part of the purchase costs budget into personnel costs. Modification of Table 3.1h with changing the participant's numbering and deleting HEK whose tot. purchase costs do not exceed 15% of pers. Costs.

Part B

Version	Changes
V1.1	General: All references to MDH changed to MDU
V1.1	State of the Art: Additional text added to improve this section.
V1.1	Impact: Further detail added here as per ESR.
V1.1	Participant Table: Updated Coordinator from TWI to MDU and Updated participant numbers in anticipation of changes to associated partners
V1.1	Gantt Chart: Deliverables 7.1 and 7.2, lead changed to MDU from TWI
V1.1	WP Descriptions • WP7: o Effort Updated for TWI and MDU o Tasks 7.2 and 7.3 lead beneficiary changed from TWI to MDU • WP1-WP7 Beneficiary numbers and efforts realigned in accordance with anticipated new beneficiary numbers
V1.1	Table 3.1c: Deliverables 7.1 and 7.2 lead beneficiary changed from TWI to MDU
V1.1	Removal of beneficiaries table
V1.1	Added ethics assessment
	Developping and specifying KPI´s role on criterion 2 impact
	Changes in Section 2.2.(page 32) : "The communiation strategy will be lead by KNEIA" instead of TWI. Changes made in the description of social networks and online presence.
V1.5	Updating in the beneficiaries numbering in the table 3.1h and removal of HEK from the table 3.1h

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1. Excellence

1.1. Objectives and ambition

BACKGROUND

The 4th Industrial Revolution/ Industry 4.0 has introduced new levels of production capacity and efficiency. Industry 4.0 enables the reduction of production costs, maintains product quality and consistency and has made mass customisation economically viable (Brynjolfsson & McAfee, 2014; Schwab, 2017).

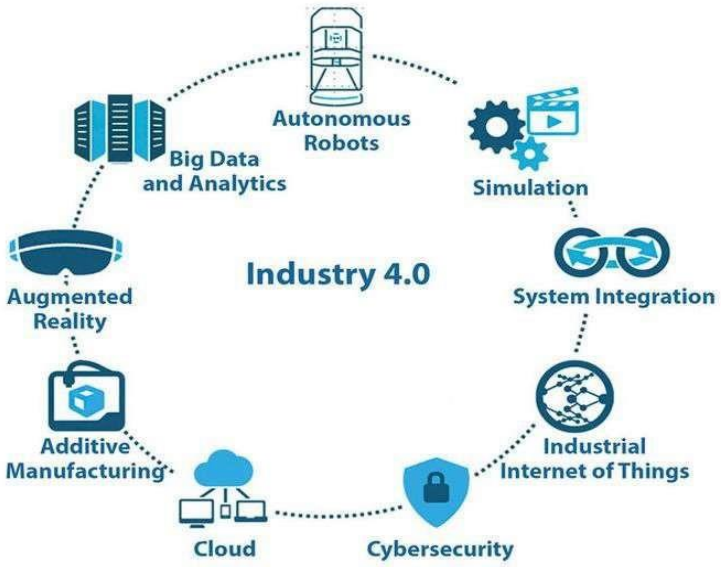


Figure 1 The Industry 4.0 Pillars

The UK Industrial Digitalization Review Interim Report (UK Industrial Digitisation, 2017) defines Industrial Digitalisation as: "the application of digital tools and technologies, in all their forms, to the value chains of businesses who make things (e.g. automotive and construction) or that are operationally asset intensive (e.g. power grids, wind farms etc.): it is the merging between the physical and digital worlds". Industry 4.0 is a general-purpose technology, one with applicability in a wide variety of settings. It can, therefore, be expected to diffuse widely across all sectors of industry with positive consequences for whole regions and economies. Industry 5.0, by contrast, is concerned with leveraging the added-value that human creativity and ingenuity can bring to increasingly capable and accurate automation technologies.

However, industry 4.0 (in particular automation) has been plagued by the perception that it will lead automatically to job cuts, wage deflation and in some sectors, the complete replacement of the human workforce (Benavav, 2020; Frey, 2021, Barley, 2020). Job replacement theory, however, shows no proper regard to the distinction between replacing and human augmenting technology implementation (Benavav, 2020, Barely, 2020). Full automation is appropriate where the only objectives centre around production efficiency, quality, accessibility and cost reduction. In contexts where innovation, flexibility and the application of craft and artisanal skills to production are either beyond the capability of automation or themselves carry value in the eyes of customers, then full automation is inappropriate. In such contexts, effectively leveraging effective working between the human workers and technology becomes the key managerial and research concern. One might go as far as to argue that, because the profits from digital technology and automation necessarily tend toward zero as production technology inevitably diffuses to all competitors, that ultimately only creative and skilled human labour can sustain an individual firm's competitiveness in the long run. It follows that sustainable competitive advantage is rooted ultimately in the

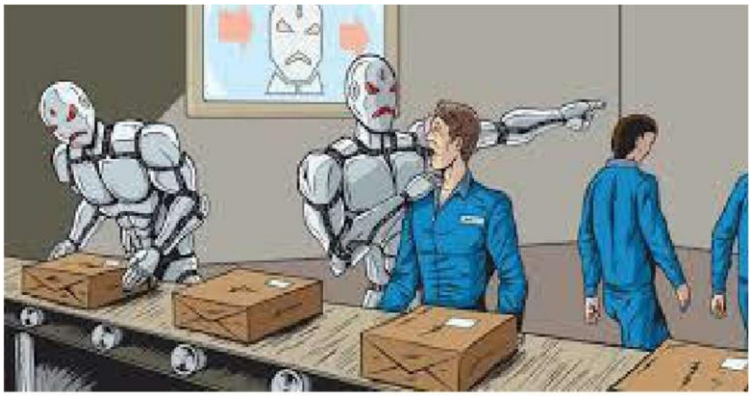
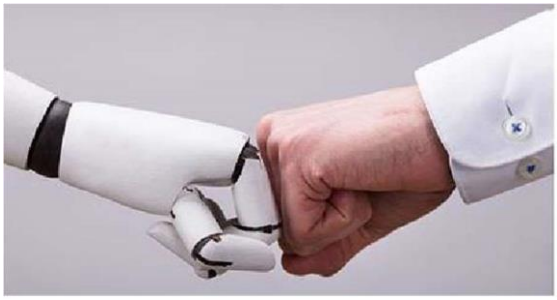


Figure 2 Automation Misconception

competencies needed to effectively manage and sustain skilled human workforces alongside automated and digitalised production processes.

AMBITION

Digital technology through sensors, connectivity, and data processing offers more refined, extensive, and remote control over processes. It allows for low-cost real-time connectivity and automation between elements of the production process (Wang et al 2015, Wang et al., 2016). The ability to access manufacturing process data in real-time improves strategic and operational decision making (Porter and Heppelmann, 2014, Schwab, 2016) and so allows production to run more flexibly, reliably, and efficiently (Shrouf et al 2014; Hozdić 2015). Thus, adopting firms are able to lower maintenance costs (predictive maintenance based on wear data), reduce down time, are able to run processes at higher capacity with fewer staff, and better respond to consumer demands for product customisation (Zawadzk and Żywick 2016). Manufactures can themselves manufacture goods with integrated digital capability (Ayala et al., 2017, Coreynen et al., 2017) which then subsequently yield user data that enable new services; potentially enacting shifts in manufacturer's business model toward service delivery (such as charging for functionality that can be turned off and on remotely).



Thus, the application of industry 4.0 technologies will, it is claimed, continue to disrupt companies giving rise to new business models, innovative products, and enhanced services (Made Smarter, 2017) and the potential for radical transformation in supply chain relations (Langley, et al. 2020; Brynjolfsson and Mcafee, 2016). As such, policy makers see industry 4.0 is essential for future growth and competitiveness (see, for example, European Commission, 2016; UK Industrial Digitisation, 2017), actively encouraging firms to pursue this as a way of fostering future growth and raise economies out of stagnation.

The benefits of I4.0 will not come to firms automatically, however, and the particular strategic context of each firm will shape how they should approach i4.0 implementation. For firms for whom the complete automation of production is not a strategic goal an I5.0 approach is appropriate. An i5.0 mind-set seeks to exploit what we term 'inter-augmentation' between workers and i4.0 technologies. The principle that pairing skilled workers and craftsmen with machines is a productivity and innovation game-changer for those firms that can master it. However, knowledge of industry I4.0 remains patchy in many areas of industry (UK Industrial Digitisation, 2017; Made Smarter, 2017; Stentoft, Rajkamur and Madsen (2017), studies of I5.0 are still in their infancy and detailed ethnographic studies of I5.0 are presently non-existent. This project presents an opportunity not only to identify the key features of the topology that defines human-machine relations in industry 5.0 but will define the managerial competencies needed to succeed in this space. The project will link varied firm strategic contexts, technology implementation and detailed workplace ethnographies in a single research project.

We address primarily the workforce implications of industry 5.0 and in particular the relationship between automation choices and maintenance of skilled work. We will thus address an under-explored strategic space in production - the zone where automation adds value to skilled and artisanal work and where further automation risks over-automation and undermining the value of what is produced.

Our research will identify how the potential for automation and human input is being played out in a range of industrial, competitive and supply chain settings by creating detailed comparative case studies. We will use ethnographic research methods (shadowing, photography, observation and interviews) to identify the ways in which

job roles and content are changing as a consequence of the introduction of i4.0 and 5G technology. Following from this, the research will identify the shifting managerial capabilities that are needed to ensure the maintenance and added-value in these spaces, such that technology implementation migrates from a substitution to an augmentation approach. Therefore, our work will also support the preservation of cultural heritage in the form of craftsmanship skills that might otherwise be lost.

OBJECTIVES

We have broken down our ambition into the following objectives:

Objective #1: Identify the nature of the different business, industrial and market contexts in which skilled workers are likely to continue to add and be valued.

Why it is important: Automation does not simply, by virtue of its inherent nature, 'threaten' jobs. Nor do jobs carry inherent irreducible characteristics that render them automatable. Different attempts at classifying 'at risk' occupations have, unsurprisingly, come to wildly different conclusions (Frey, 2020). Our research will focus on the strategic, social, economic and business contexts in which it does or does not make business sense to automate skilled, artisanal and craft roles.

How it will be delivered: Value stream mapping, linking to contextual factors, of skilled craftsman roles threatened by automation in each business context identified/used

Objective #2: Identify how skilled and creative workers act to ensure their continued place in the value chain.

Why it is important: Workers are not passive recipients of technology change, they also react to it. The history of technology change is replete with job roles that have proven unexpectedly resistant to elimination by automating technology (Barley, 2020). Railway signalman, for example, despite signals themselves being automated, maintained the existence of their role by expanding it to take on new tasks (such painting and welding). Recent experiences of digital technology and AI roll-outs also make clear that new human roles constantly emerge to augment technology that rarely works quite as well anticipated (Crawford, 2021).

How it will be delivered: Identify individual, community, and organizational practices that enable role survival and adaption in the face of automation.

Objective #3: Identify, and detail, good practices in organisations that continue to mobilize human skill, ingenuity and creativity as a source of value and competitive advantage.

Why it is important: We lack understanding of precisely what managers and organizations do to better lever benefits from interacting human and technology resources. Brynjolfsson & McAfee (2014) identify the importance of distrinbuted decision making for taking best advantage of digital tecchnology, however we know little if anything about the detail.

How it will be delivered: Good organizational and managerial practices identified.

Objective #4: Examine the specific organisational and managerial competencies needed in contexts where skilled and creative roles meet autonomous, semi-autonomous and intelligent systems.

Why it is important: Before the introduction of automation systems in a skilled crafts environment, both the organization and the managers require a certain level of skills to both communicate and implement the new approach. Therefore, there is need to carefully audit all the requisites at organizational and managerial level for the specific level of automation that will be implemented.

How it will be delivered: This is expected to be industry specific hence a case study approach will be use to develop the general guidelines.

Objective #5: Develop pathways for integration of I5.0 and craft skills for different case studies

Why it is important: This will provide the specific guidelines that can be used in the introsuction of I5.0 to the European manufacturing sector.

How it will be delivered: Integration Guidelines and troubleshooting references

Objective #6: Test the developed pathways in related companies

Why it is important: This will allow us to verify our claims and potentially quantify the expected benefits for a better value proposition and establish a willingness to pay for the pathways

How it will be delivered: Demonstration workshops for the respective companies

Objective #7: Develop and carry out proper dissemination, communication and exploitation strategies to maximize the beneficial impacts of the project outcomes, ensuring a successful scaling-up and market uptake, to boost European industry competitiveness.

Why it is important: This wil enable us to get the idea to as many European companies as possible. It is relevant to keep as many of the skilled Europeans working as possible for the embitterment of their lives and maintain the European way of life as well as in compliance with the SDG#8 Decent work for the citizens.

How it will be delivered: Participation in industrial conferences and exhibitions on I5.0. Publications in journals

Objective #8: Design of management training course content and course modules for I5.0 introduction for the management and the workers

Why it is important: The successful future of I5.0 relies on the training of both workers and management to ensure that the benefits are well extracted. This is aimed at ensuring there is a place for skilled crafts within manufacturing while achieving production competences such as high throughput, quality and product consistency as can be achieved using I4.0.

How it will be delivered: Training modules that can be implemented in isolation at the production plant or as part of an advanced manufacturing course or even MBA

STATE OF THE ART

The following table summarises the general state of the art about some existing methodologies in use or under development that relate to our project and how Up-Skill might provide benefits. Table 1 State-of-the-Art Analysis

Alternative solutions in use or under development	Limitations (What is it unable to provide?)	How our solution is better (quantitative and qualitative assessment)
Replacing skilled labour with automated processes	☐ De-skills workforce ☐ Is disruptive and meets resistance ☐ Demotivates and creates an unemployment problem	• Up-skills workforce • Is welcomed and accelerates the uptake • Is motivational and reduces an unemployment/re-skilling problem, particularly in the light of COVID
Increasing traditional training programme	☐ Is too slow to enable a reactive approach to demand Very ☐ high cost	• Is rapid and responsive through the use of data • Is low cost

Reduction in high-value production volumes	Loss of quality in production Tendency to offshore to low-cost production centres Loss of skills and products which retain cultural traditions and enhance the quality of life	Ability to retain skills and cultural traditions in a modern context to enable viability.

This section refers to the now well established dichotomy between upskilling and deskilling in the literature on technological change and skill. It is well established that the worst predictions of technological unemployment, most recently popularised through discourses around automation and the 'end of work' (Benanav, 2020, have not come to pass. The evidence is less clear on how skills have been impacted, with some digital technologies having a negative

impact on pay and conditions, and breaking unionized controls over skill (Woodcock and Graham, 2020), whilst others have been disruptive, but created new, often highly skilled jobs, though not necessarily going to the same workers. The broad consensus revolves around variations of the 'polarization thesis' that technologically led innovation generates a combination of new, highly skilled roles, combined with deskilled work and unemployment (Lopes and Calapez, 2021; Martinaitis et al., 2021). The higher-skilled roles rarely go to existing employees, leading to a disparity between the aggregated statistical outcomes, and individual or firm-level experiences. All of these approaches depend upon relatively static conceptions of skill determined by prior skill taxonomies in the industry, or qualifications, neglecting the ways in which skill itself is socially constructed and contested through organizational change processes (Warhurst et al., 2017). This allows a more nuanced understanding of skill in practice and in context, allowing a recognition of skilled work that is often neglected, at great cost to companies, a less expensive and rigid approach than large-scale retraining programmes, and a maintenance of significant, strategically and culturally valuable skills, in a modern, digitally mediated business context.

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1.2 METHODOLOGY

The focus of this project is to develop a better understanding of how businesses, particularly in industrial/manufacturing environments, can lever value from human and machine integration. Businesses are already exploring how they can build business benefit from newly added capabilities such as big data, access to real time process data, IoT, automation, machine learning and artificial intelligence amongst others – the next step will be integrating those capabilities with human capabilities (such as creativity, ingenuity, artisanal skill).

Studies of digital technology in the workplace have typically followed what Barley (2020) refers to as an isolationist approach - seeing roles as either replaceable or not. Thus, for example, Woodcock & Graham (2019) have referred to 'digital legibility' when determining whether a role is likely to be illuminated or not. Frey and Osbourne, in theirnow famous account of what roles are likely to be eliminated, used the content of the role to assess whether the role was likely to survive or not.

Barely (2020) has suggested a more wholistic approach, one which instead looks at how the tasks associated with particular roles move between roles or even create new roles. Thus, we should not look at roles in isolation but a) look at the network of roles of which the role is a part and b) we should focus on the actual tasks associated with those roles and how they alter in relation to technology change. Barley (1984) talks about technology change reverberating up through the organization, from role to role and potentially, into the permanent structures of the organization itself.

Alongside a focus on roles, role networks and tasks we propose two further methodological iterations. The first brings the role of the manager more sharply into focus. Not only does the manager and the management function play a crucial role in how technologies are used by workers in practice, but they may also play a key role in selecting the technologies that are implemented within the organization in the first place. Secondly, and relatedly, we suggest a need to account for the broader strategic context in which the technology implementing organization exists. Thus, the markets in which the organization operates may prove to be a crucial factor in deciding where the balance between automation and craft skill lies. For example, in some markets clear evidence of human intervention will be crucial to the value of the product or human intervention, such as in welding, may remain a practical necessity if the organization wishes to remain in markets that demand low-volume, varied and bespoke output.

We thus also go beyond also existing ethnographic research concerned with technology and work by fully taking into account the business and strategic context of the organisation and link these, through managerial competencies to operations and the detailed organisation of work. While contemporary ethnographers of work and technology acknowledge structural, power and cultural differences between organisations and the importance of identifying how consistent the effects of a particular technology are across different organisations (Barley, 2020) neither he, nor his

colleagues in the field of work and technology ethnography, (e.g. Orlikowski, 2007; Leonardi, 2013; Orr, 1996) pay attention to either business contexts or the role played by shifting managerial competencies.

In our view, the business reasons for the adoption of any technology is critical to understanding how it is adopted. Associated with document Ref. Ares(2022)4648166 - 24/06/2022
Thus, for example, Foxconn, the Apple Iphone assembly giant, has its workers sitting around tables assembling Iphones by hand from piles of components lying on the table. The automation technologies that could do this work are well established – and would barely count as innovation, but they are not adopted because they do not support its business model. The reason for this is partly low-wages, but it is also the specific business context in which the firm operates. To keep its products up-to date and allow last-minute changes Apple prefers the production flexibility of human hands (Freeman, 2018). Put another way, the precise nature of the business space in which humans meet automation depends on the broader business context as much as local organization and work-culture.

We will also seek to provide a clear pathway to integrating skilled craft and automation by identifying the pivot which ensures that a business can harness both. This will involve dissecting the roles of skilled crafts and determining the level of automation that can be included in the appropriate positions. To undertake this, we will account for the context of each business through investigating what exactly the company does, its value proposition, its competitors, the nature of market it is in, the ways in which demand for its products is structured. We will then focus on the skilled craftsmen as the human component that adds value to outputs as automation usage increases in different organizational and business contexts. Therefore, we will conduct our research with both managers and skilled / artisanal workers. The full scope of this work will enable us to develop I5.0 integration pathways for businesses.

Figure 4 below summarises the overall steps we will use in this project to determine the input information that will allow us to develop the aforementioned integration pathways.

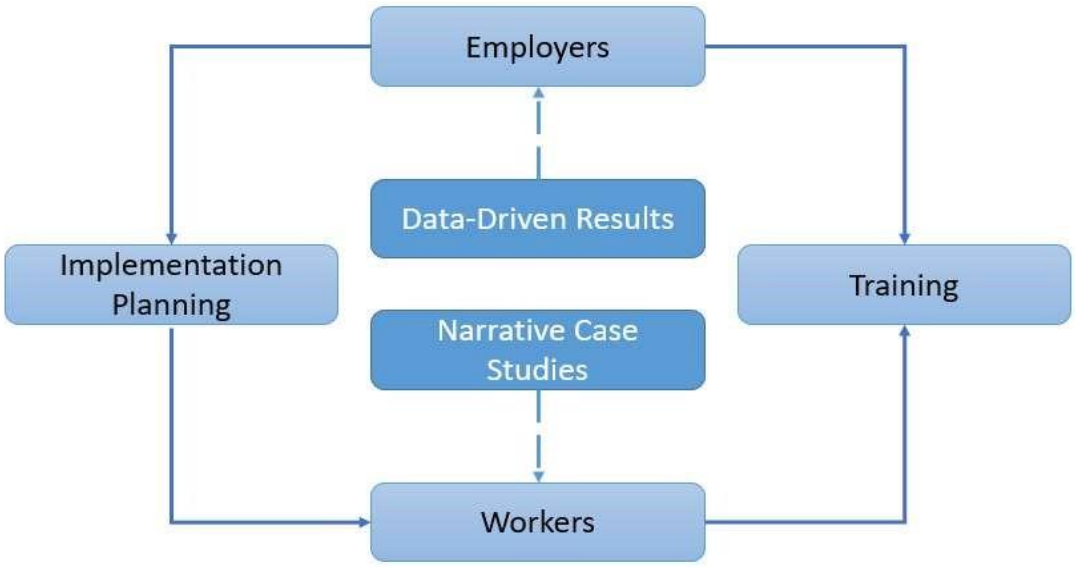

Figure 4 Methodology Summary

To deliver on our objectives, our approach will centre around the use of detailed ethnographic in combination with the application business analysis tools and techniques.

Ethnographic Research Techniques

It is important to see how work is actually done to assess how it is changing. It is notoriously difficult, for example, for workers to describe in interviews precisely how they do their work and what their skills are (Barley, 2020). It is also essential that researchers understand, as far as it possible, the technologies that shape the workplace. Similarly, business managers and leaders are often are not best placed to comment on the work of others, as the various business and performance drivers they are mindful of can prohibit clear, authentic, impartial assessment.

We will deploy a team of researchers for the ethnographic analysis, with one ethnographer dedicated to each - industrial case region for the 12 month duration of the research period. Each researcher will be embedded in the

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respective use cases utilising a combination of shadowing, photography, document collection, observation and interview. During the 12-month use case investigation period, they will carry out:

- Role Change Analysis; this aspect of the research will identify the tasks undertaken skilled and artisanal workers; what is it now and how the role changes with the introduction of new technology?
- Role Network Analysis focuses on understanding how tasks are re-distributed differently between different roles when new technologies are implemented as well as how communication patterns, loci of control and responsibility change. Also, what is taken on by others? What new roles are created?

Role Change Analysis

In our approach, we are seeking to clearly define the full breadth of what each role a skilled craftsman carries out entails. This will include identifying non-routine tasks, the level of autonomy of the role and continuous skill building capacity.

In role change analysis, it is key to identify all the non-routine tasks in the role of a skilled craftsman. These tasks are defined as the extent to which work relies on the variety of non-repetitive tasks that cannot be easily monitored or codified. We will also investigate the shifting level of autonomy of different roles. – that is, the extent to which workers can decide precisely how and when to perform different tasks. This will involve also identifying the opportunities for workers to learn new skills and to pass on skills through training and mentoring.

Figure 5 shows how a role can be constantly improved and transformed from delivering an intangible output to a tangible output after role change analysis

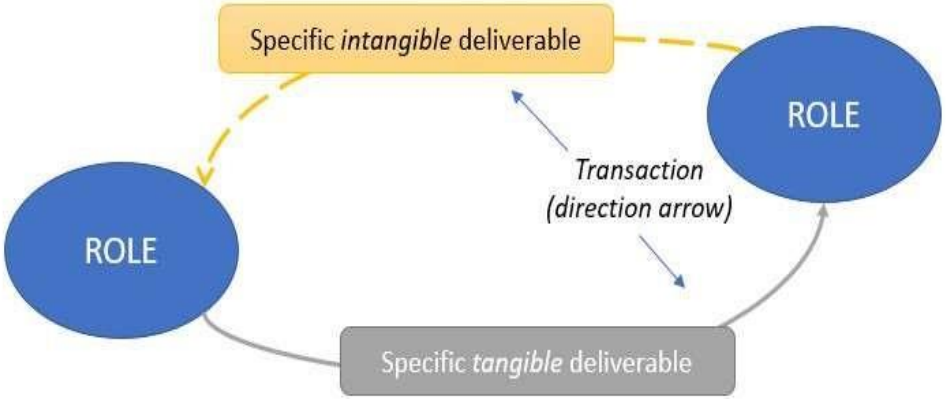

Figure 5 Role change dynamics

This is key in mining each role to maximise its full potential by ly positioned interventions.

Role Network Analysis

This seeks to understand technology affordances and how they will enable the transference of tasks from one role to another. The idea is to clearly define the key benefits of the technology at the company level as well as understand how it changes the interactions between co-workers and departments. This will also aid in defining what change brings value to the company as well as identify areas where the skilled craftsman requires training if the role now has new features. This analysis helps identify any dis-benefits of the new technology as well as any benefits that have not been identified and realized within the business. Figure 6 shows the matrix of machine affordance and perceptual information.

- Correct Rejection implies where automation has not been used at all and based on the researcher's analysis these areas indeed do not need automation
- False Affordance identifies areas where automation has been used and assumed to have benefits but does not.
- Perceptible Affordance identifies areas where the benefits are clearly visible and accurately apportioned
- Hidden affordances represent areas where automation can provide benefits but users are unaware.

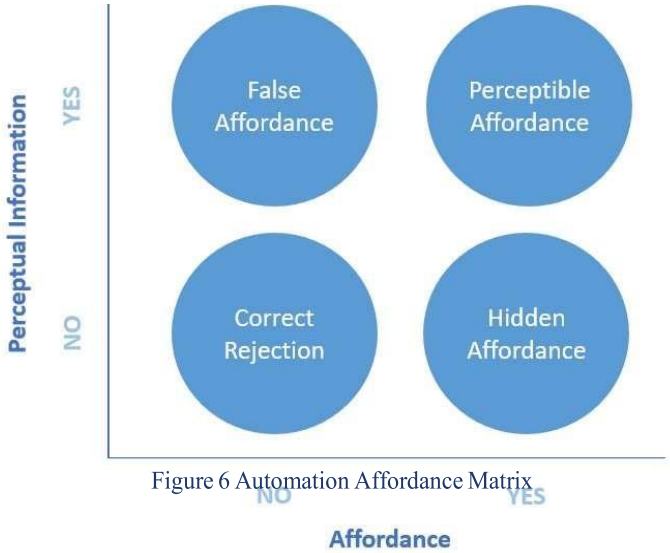

Figure 6 Automation Affordance Matrix

The idea of this analysis is to elevate all hidden affordance to the perceptible level while pulling down all false affordance into the correct rejection sphere.

Business analysis and change methodologies

The researchers will also have the following tools available to them, to supplement and strengthen the data collection approach while also increasing the value of the research to the case study organisations:

Business Model Canvas

We will use the BMC to identify how precisely the business creates value in the markets in which it operates. How value is created will have a direct bearing on how skilled work is perceived. So for example, skilled workers may be various understood within the business as a cost or an integral means of adding value. Automation and othecontrol systems may be seen as a means to help retain skilled workers (undertaking heavy or unpleasant work) and so as a means of attracting and retaining them or as a means of de-skilling, controlling, externalising or eliminating them. To help identify how the business extracts value from what it deems high-value skills as well as around what they rank lowly. This will allow us to dissect the impact of the skills on the overall business. This is key as often management is usually oblivious of their most valuable skillset.

Technology Benefits Identification

Technology implementation is always a risk to the business. Technologies in the form of capital machinery are (often) expensive investments and often do not operate as intended (Mindel, 2015). This will be implemented as a workshop that identifies the implications of particular technology choices while identifying the requirements of organisational change to extract benefit. Workshops last half a day and involve managers as well as those working

on the shop floor. Discussions cover the risks posed by the technology as well as the need for other actions like training, task restructuring or organisational re-structuring. The relationships between technology enablers, business changes, benefits and organisational objectives are mapped visually to foster open disucsion.

LEAN 6 Sigma

Lean thinking and analysis tools will be used to refine and improve how existing technologies add value to the business by identifying waste stemming from the present and planned set up of operations with a particular focus on integration with skilled craftsmanship and the total utility of the available automation capacity. We will investigate redundancies for both human workers and machines, identifying where either of them can better take control, step aside or collaborate to eliminate time and resource waste.

Change Management Support and Analysis

We will use our knowledge of the change management process to address potential problems that will arise as companies seek to transition to I5.0. This will include developing communication methodology to explain

- why the change is necessary
- what new values the individual need to embody
- what are the expected benefits from each individual's value shift
- what guidelines need to be developed to maintain the new value system for the company
- What are the necessary steps that will enable the realisation of the desired benefits

Outcome Categorisation and case study selection template

The outcomes we expect to see from the project fall within a sliding scale of impact from automation through to augmentation and digital feedback as defined within four broad categories which we expect to identify and refine through the course of the project. In each category, we are targeting at least one case study. The categories outlined below also give us a sense what the orginal strategic intent was in implementing any given process technology. It does not of course follow that the orginal intent of any given adoption strategy will be played out in reality. We recocognise that AAA ,in table 2 below, is a strategic intent that is rarely realised entirely in its purest form.

Table 2 Outcomes Categorization

CODE	CATEGORISATION	DESCRIPTION
AAA (e.g., Offshore wind turbine repairs)	Full Automation	It is likely the job will ultimately cease to exist because it can be wholly automated, as and when equipment is available and becomes a cost-effective option
AAH (e.g., Ship Building)	Irreducible Human Integration	Roles will be largely automated but require certain human intangible characteristics/intelligence/judgement which cannot be automated, and perhaps should not be automated
HAA (e.g., Ornate Shotguns and fine watches)	Strategic Human Integration	These are Industrial-scale products, but must retain a level of skilled craft to maintain the quality of product and desirability to consumers
HHA (e.g., Gold leaf Augmented with technology, where appropriate, to augment their craft and manufacturing) Technology maximise its potential market value and deliver efficiency	Artisanal work	A skilled craft, often an isolated artisan who seeks to engage

Case study selection

Webber – Webber, located in Germany, manufactures gas burners and gas-air mixing control systems for industrial uses. The manufacturing process up until now involves a lot of human skill (for instance silver soldering of fine precision tubes), which in these modern times is not part of the apprentices curriculum anymore. At present the company relies on skilled technicians to assemble burners, but their ambition, as they struggle to replace retiring staff, is to fully automate their manufacturing activity. Their ambition, at least, falls under the AAA category.

Alfa Laval, located in Sweden Alfa Laval manufacture separators and associated components. The firm, in that it undertakes a variety of engineering quite different activities (different markets, different volumes) falls under a number of categories. However, they are keen to advance their level of industry 4.0 capability and automate where they can. We therefore locate Alfa Laval as AAH.

Ford, located in the UK has launched multiple Industry 4.0 initiatives to support their business from the initial installation to the ongoing running of operations. However, there have been difficulties due to human reluctance to adopt and work alongside technology mainly due to a lack of trust. This issue also extends beyond operations and into manufacturing planning where this case study is focussed. FORD have an automated method utilising AGVs to undertake the floor outlining process generating data and eradicating the risks incorrect install location. Additionally, post-install verification would be undertaken using a laser scanner onboard the AGV to collect data to precisely cross-reference against the planned location. However, PTME engineers are reluctant to move away from the current manual process due to a distinct lack of trust that the I4.0 technology could be incorporated successfully. This case study, would look to leverage the skills of the consortium to develop a suitable AGV based system that could overcome adoption reluctance and lead to lessons that could subsequently be deployed to improve the human AGV interface across Ford. Therefore, this project is appropriately situated at the cross-section of I4.0 and social sciences and is AAH.

Helekopter, located in Sweden, offers engineering and business services to the engineering firms in its network. Their business model is to take on the administrative and financial burdens of those firms, using the latest management software systems, and so allowing those firms to focus on value-adding 'engineering' work. Helekopter does not seek to replace engineering workers but, in effect, to allow them to out-source non- engineering work. Helekopter thus reflects our HAA category, in which skilled human work is retained and augmented through industry 4.0 technology. We will conduct research with Helekopter but also with appropriate firms with which they work as opportunities arise.

Milan (Italy) region cases. Our Milan University partner will enrol a small number of small and medium scale companies involved in styling, designing and prototyping industrial design products in a diverse range of sectors including furniture and furnishings, interior design and yacht design. The researcher responsible has relationships with such as companies like Cassina, Edra, Anonimia Castelli, Benetti. The work performed by employees in the case study organizations involves powered machinery in the making process in addition to hand tools. Although based on large scale production, the making process also involves a certain level of skilled craft to maintain the quality of product and desirability to consumers. Production still relies on direct human interaction and engagement with raw materials and manufactured products. We anticipate these cases will populate the HAA and HHA categories.

RELEVANT NATIONAL AND INTERNATIONAL RESEARCH

This project will build, where possible, on results of previous national and international research and innovation activities. The project will mainly take advantage of previous EU funded research activities on Industrial 4.0, Impact of technology on jobs business models and welfare, Automation and Work-Study, components performed in the following projects; - GrowIn 4, Beyond 4.0 and KYKLOS 4,0. Our project will also contact and utilise the developments of the EU funded project HuMan Manufacturing which is working on demonstrating ways factory workers and automation can operate in harmony, increasing productivity, quality, performance, satisfaction and

safety. Table 3 summaries all the key projects and research activities linked to the Up-Skill project and where consortium partners were involved

Table 3 Previous Projects

Project	Partner Involved	Relevance
5GEM - 5G Enabling Manufacturing UK (DCMS – UK)	TWI, FORD, LUN	Utilising 5G sensors for data capture and analytics in manufacturing environment. The use of 5G in manufacturing to connect machines allowing real-time feedback, control, analysis and remote expert support. Two mobile private networks are being installed in Ford's Dunton facility and TWI Cambridge, respectively. Ford will focus on the connectivity of welding processes, used in the manufacture of electric vehicles. TWI will support Vacuum Furnace Engineering in connecting their heat treatment equipment. LUN will work on the associated challenges such as cyber security, scale-up, standards, machine learning and developing the 5G technology for an industrial environment.
GrowIn4.0	ARU, TWI	Examining and supporting the uptake of digital technology amongst manufacturing SMEs in the North Sea region. Developing business models and tools. August 2017 to September 2021.
The Digitalization of Management. Funded by the Swedish national body FORTE.	ARU, MDU	The project is examining changes in management practice as a consequence of digital technology – a comparison of Sweden and the UK. January 2017 to Dec 2022. Collaboration between ARU and Malardalen University, Sweden. The aim of the program is to understand the impact of digital technologies on the way organisations and work are managed, and the way organisational practices impact on how technologies are enacted. Different technologies are analysed, from automatization to virtualisation of work.
DigiConCo-Op	ARU	Transforming Construction Network+. DigiConCo-Op: Transforming Micro-Project Delivery through Digital Co-Operative Construction. ESRC CN+. Feb 2020 to July 3rd 2020.
ROSSINI (H2020)	IRIS	Project Purpose: Robot enhanced Sensing, intelligence and actuation to Improve job quality in manufacturing. IRIS's role is to develop a digital platform and AI algorithms
SHAREBOX (H2020)	IRIS	Secure Management Platform for Shared Process Resources, coordinator
Weldzero (Innovate UK)	LUN, TWI	AI and MI used to analyse data to ensure defect free manufacturing
RESURGAM	TWI, LUN	H2020 – Industry 4.0 platform with MI and AI for control of FSW
FSWBOT	TWI, LUN	Innovate UK – AI and MI to detect defects and repair.
SafeFlame EU FP7 Project	TWI, WEB	Burner development for use with hydrogen for a hydrogen welding set.

The Organization of Craft Work	UMIL	This project investigates aspects associated with the future of work and the impact of technological developments, such as in robotics and computing, that appear to threaten to eradicate the need for human judgment and interaction in many jobs. The project also explores the role of technology in craftwork organizations, the relationship between the body, the crafted object and the mind, and the importance of skill, knowledge and learning in the making process

Craft Imaginaries – Past, Present and Future	UMIL	This project investigates the social imaginary that surrounds modern craft. It focuses on the relationship between the craft imaginary and authenticity that is related to the temporal character of craft in constructing past–present–future relationships. The inherited value of craft imaginaries resides in possibilities to build from the pasts and construct meaningful translations in the present and envisioned futures
Several automation projects at our tech centre with partner companies	MITC	Intelligent robotisation and automation projects in production processes and specific cells with eg Alfa Laval, GKN, including vision, sensors etc
Several projects in visualisation and simulation in production	MITC	Using e.g., HoloLens 2, Visual components, AI in the production process. E.g., impact on operators work environment

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Gender Dimension

The consortium partners are committed to the active promotion of gender equality and will fully comply with the European Union policy of equal opportunities between women and men. The partners involved in the proposed research operate equal opportunity policies concerning staff recruitment and deployment and will contribute to promoting gender equality by continuously encouraging and supporting women's participation in all the areas of the project: research, development and industrial application of integration pathways to be produced at the end of the project. With this in mind, the project coordinator will assume responsibility for ensuring the participation of women at all levels of the project. In the consortium, some key personnel involved in the project are women. The project will make the best possible effort to improve the gender balance and raise awareness of the equal working opportunity for both women and men.

Open Science Practices

All partners are committed to publishing the knowledge generated (as the main route of the dissemination) in scientific, peer-reviewed open access journals as well as in national and international conferences. In line with the principle of Horizon Europe explained in the Regulation and the Rules of Participation as well as through the relevant provisions in the grant agreement each beneficiary must ensure open access (free of charge, online access for any user) to all peer-reviewed scientific publications relating to its results. We shall deposit an electronic copy of any published version or the final manuscript accepted for publication of a scientific publication relating to the foreground before or after the final report in an institutional or subject-based repository at the time of publication. Beneficiaries will make their best efforts to ensure that this electronic copy becomes freely and electronically available to anyone through this repository: immediately if the scientific publication is published "open access" (i.e., if an electronic version is also available free of charge via the publisher) or within 6 months of publication. All the future exploitable activities of the research outputs will be done under fair and reasonable conditions and will commit to rapidly and broadly exploit the resulting products and services, as described in the Data Management Plan as well as in WP5 (Dissemination, Exploitation and Communication).

Data Management

Data generated during the project is anticipated to be in the form of research/ feedback reports covering both mechanical data and socio-economic data. A detailed data management plan (DMP) will be produced within the

first six months of the project. This will enable the collection of data generated throughout this project and will be essential for achieving any regulatory approval. Compliance with ISO standard ISO 15489-1:2001 information and documentation – records management, as well as the General Data Protection Regulation (GDPR) about the protection of personal data, will be ensured. All research data generated during the UP-Skill project will be managed following FAIR principles to ensure that it is Findable-Accessible-Interoperable-Reusable. Research data will be deposited in a 'trusted' repository and access provided under the principle of "as open as possible, as closed as necessary". The consortium will also comply with additional Horizon Europe requirements concerning the

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bibliographic and administrative metadata that relates to deposited data. Confidential data (test methods and results) will only be shared within the consortium until the protection of the intellectual property is secured. Non- confidential information will be shared more widely at conferences and trade fairs. The data will also be stored on secure servers with remote backup, internal to the consortium members under the controls specified in their existing ISO9001 quality management systems. All data that is uploaded into the repository will be in the machine- searchable format and made freely accessible to other researchers using Creative Commons Attribution Licenses (CC BY), which spell out terms of use and how the data can be used by others. The data will be held for a minimum of fifteen years after the end of the project.

The DMP will cover:

- Types of data/research outputs: definition of dataset units to be produced during the project life span. Each data set will contain, at least the work package to which it belongs and data typology, with a short description.
- Findability of data/research outputs: datasets will be stored on Zenodo, automatically becoming part of OpenAIRE, the EC-funded initiative supporting the Open Access policy. Each will be assigned a unique identifier.
- Accessibility of data/research outputs: use restrictions and licensing information. Open Access will be implemented in peer-review publications, conference proceedings and workshop presentations.
- Interoperability of data/research outputs: standards used to describe datasets. Proper metadata facilitates the use of the data by others, makes it easier to combine information from different sources and ensures transparency. Metadata of deposited data will be open under a Creative Common Public Domain Dedication (CC 0).
- Reusability of data/research outputs: the right to access and reuse digital research data under the terms set out in the Grant Agreement will be ensured. The DMP will include information on licensing of data, their availability, re-use of data, duration of data for re-use and reproducibility of research outputs, among other issues.
- Curation and storage/preservation: The Exploitation manager will oversee data management, while quality control of each dataset is the responsibility of every WP leader. Informed consent in data collection and information protection in data storage and access will be considered, as well as procedures for long-term preservation of datasets.

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2. Impact

2.1. PROJECT'S PATHWAYS TOWARDS IMPACT

Automation, digitalisation and other related innovations are expected to significantly change the nature of work. In future there is likely to be a surge in demand for some skills and a reduction in demand for certain physical and manual tasks. Understanding these shifts will help society, policy makers and business leaders find ways of maximising the benefits while recognizing and mitigating related challenges. The key challenges and opportunities created include.

- Talent is currently underutilised and labour markets are under strain – according to McKinsey, over 285million adults are not in the labour force in the EU-15 countries and the US alone, and more than 100 million of them would like more work1. Approximately 850 million (30 – 45%) of the working-age population across the US, UK, Germany, Japan, Brazil, China and India, is underutilised2. At the same time, over 75 million youths are unemployed. Moreover, over 655 million working age women are less economically active than men. Improvement in skill capacity utilisation across the board will contribute an additional €12 trillion annually to EU-GDP by 2025.
- Many activities that workers carry out today have automation potential – MGI research on automation potential3 estimates that the automation potential for current occupations is less than 5%. Approximately 60% of all occupations have a 30% technical automatable potential, benefiting highly skilled workers while enabling more productivity for low-skilled workers working with technology4. Globally (China, India, Japan and the US alone) it is estimated that automation could affect over 50% of the global economy, across 1.2 billion employees and €14 trillion in wages5.
- Automation has the potential to help labour markets – Digital talent platforms have emerged over recent years and are intended to improve ways in which workers and jobs are matched6. Online screening platforms speed up the hiring process while cutting time individual spend searching for jobs. They have the potential to address geographical mismatches, putting the right people in right jobs thereby increasing productivity, however at present there are serious limitations within the keyword matching algorithms, resulting in a tendency to select out potential workers with transferrable skills.

The ethnographic approach will enable us to build a unique and detailed understanding of the ways in which artisanal skills and automation interact and are managed in each of our cases. Precise ethnographic methodological design will be critical here in order to build detailed understanding of the products made, the technologies used to produce them and the specific human skills (both tacit and explicit) deployed to produce them (Barley, 2020). Our approach will also take specific account of the firm strategy and managerial competencies that are delivering business sustainability and growth. The ethnographic design will highlight:

1. The specific nature of the production technologies in use and the products being produced
2. The specifics of the supporting infrastructure and inputs relating to production as it is presently enacted. What is termed the 'technology stack' (Barley, 2020)
3. The specific skills required at all points in the production process and how they are distributed and combined across different roles. These would include technology-based as well as social skills (such as work planning, innovation, problem solving).
4. The managerial competencies needed to bring both the material and skills elements together in the most effective assemblage or functional machine (Suchman, 1987) involving both artisanal skills and automation.

The pathways to impact will delivered through the provision of a detailed understanding, for each case study organisation, not only of the specific skills required to deliver effective production within a specific strategic business context, but also what managerial competencies and firm structures are needed to ensure the firms competitive position and USP.

Collectively, our work will inform the precise skills and characteristics needed by both firms and workers in a way that is contextualised in the actual conduct of the businesses we deal with. In combination, the cases offer a sufficient broad range of organisations as to be able to generalise about how we deal with skills in relation to automation - not as isolated 'possessed' attributes of individual workers but combinatory elements of entire socio-material production processes.

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It is not straightforward to quantify the overall impacts, not least in those cases where there is no counterfactual point of comparison. However, even a small benefit to European countries, reducing the potential of automation to generate technological unemployment and under-employment, would be significant. Widely cited estimates up to 47% of employment are in a 'high risk' category and 'potentially automatable over… a decade or two' (Frey & Osborne, 2017), suggest that even a small impact on automation through skill recognition and development in training will significantly reduce technological unemployment. Given a working population in the EU of 188m (2020 statistics), assuming a similar impact to that identified by Frey & Osborne would mean that up to 88.4m jobs were at risk of automation in the near term. Even a 1% reduction in that figure would improve over 884,000 jobs, reducing the potential for automation and replacement. If we review just the manufacturing sector, where automation risks are particularly high, the EU has more than 33.2m employees in the manufacturing sector suggesting at least 15m of these are at risk of near-term automation, with a potential benefit to 150,000 employees directly.

UNIQUE PROJECT RESULTS CONTRIBUTION TOWARDS OUTCOMES SPECIFIED IN THE TOPIC

Up-Skill will have wide reaching impacts which directly affect the evolution of Industry 5.0 and its implications for future work, and in turn European and worldwide industries and policy.

The top-level outcome we expect to see from the project is the development of a quantified, predictive approach to the implementation of Industry 4.0 principles within manufacturing industry (in its broadest sense). Much work has been undertaken on the technology driven principles of Industry 4.0 (developing sensors, interpreting data etc) but comparatively little work has addressed the question of how to tailor the approach to a particular company based on an assessment of local needs.

Based on our analysis, we have established that approaches fall within a sliding scale of automation and digital feedback as defined within the principles of Industry 4.0, from Substitution at one end of the scale (the role can be fully replaced by machine) through to Augmentation at the other (the human creative input defines the role but may be assisted in productivity by the use of machines without impacting the fundamental nature of the work). These have corresponding impacts on the needs for future roles. We have defined these within four broad categories which we will further develop and refine through the course of the project.

(a) AAA – Full automation: the job will ultimately cease to exist because it can be wholly automated and replaced, as and when equipment is available and becomes a cost-effective option.

(b) AAH – irreducible human integration: roles will be largely automated, but require certain human intangible characteristics/intelligence/judgement which cannot, and perhaps should not be automated. The business

1 McKinsey Global Institute Technology, jobs, and the future of work

2 Ho w advancing w o men's equalit y c an add $12 tril lio n to glo bal gr o w th

3 Harnessing automation for a future that works

4 Four fundamentals of workplace automation

5 The Countries Most (and Least) Likely to be Affected by Automation by Michael Chui, et, al.

6 Connecting talent with opportunity in the digital age

should cease to try and automate where it is counterproductive to its interests in delivering to its USPs, e.g., by

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reducing flexibility or the ability to extrapolate between qualitative data sets, and in so doing to attach value to the role of the individual craftsperson.

(c) HAA – Strategic human integration: guns, watches etc… These are Industrial scale products, but must retain a level of skilled craft to maintain the quality of product and desirability to consumers. This often means introducing automation to the extent acceptable by the market in manufacturing at the PARTS level, but the finished product is and must be assembled by the skilled craftsman in order to retain the desired quality and desirability of finish. There is often a tension found here between the craftsman and the business, relating to the definition of the appropriate boundary between 'machine made' and 'handmade' which will be explored within the Up-Skill project.

(d) HHA – Humans, artisans who see the value in using technology: a skilled craft, an isolated artisan who seeks to engage technology, where appropriate, to augment their craft and maximise its potential market value and deliver efficiency. This becomes critical to ongoing viability of a craft business, which has implications both for the individual craftsman and for retention of heritage skills where without automation or business potential the skills will be lost due to failure to attract future trainees, with the loss of both the cultural heritage and the product within the supply chain.

OTHER IMPACTS NOT MENTIONED IN THE WORK PROGRAM

The project will deliver key improvements in the partner's capabilities:

Partner	Innovation Capabilities
TWI	Cutting edge research to industry and exploration of the latest advanced materials, quantum technologies and nanotechnology. TWI's deep and unique history allows its experts to balance traditional techniques with modern technologies, to find the sweet spot between digital and physical engineering.
MDU	Innovation and product realization connected to the Division of Product Realization. Research is of a cross-disciplinary addressing several areas, including Industry 4.0, Smart Manufacturing, Cyber-Physical Systems, Internet of Things, Artificial Intelligence, Modelling & Simulation, Predictive Maintenance, Digital Twins, and Cyber-Security.
MITC	Manufacturing, industrialisation, and product development with digitalisation and sustainability as cross-sectional areas. Experts in advanced robotics, cyber-physical systems and artificial intelligence.
ARU	Through its Innovative Management Practice Research Centre (IMPact), are part of 'Grow-In 4.0', contributing to helping smaller firms to stay competitive in the digital future. The Business School at Anglia Ruskin University works together with the Faculty of Science and Engineering to lead the study on digitisation in manufacturing SMEs, and supporting business strategy tool development and SME engagement.
UMIL	Expertise in organizational behaviour and organizational studies including the nature of contemporary workplaces and the ambivalence surrounding job transformation and alternative forms of work arrangements. Knowledge on future skills requirement and contribution to systematic investigation of working practices associated with craftsmanship in contexts characterised by automated manufacture of some parts.
HEK	The optimisation and installation of a new business management system will greatly improve productivity as the company grows and optimise the interface of responsibility.

LUN	Through the Work Foundation, they back BAE Systems encouraging concerted and co-ordinated effort by industry, Government and the education sector to ensure the UK can fully benefit from the digital revolution and Industry 4.0. Supports the working together of aerospace, engineering and manufacturing sectors and prioritise investment in digital and 'soft skills', upskilling and retraining and supporting supply chains and SMEs, to respond to the expected levels of complexity in industrial and business systems.
HEK	Industrial design and development to prototyping and contract manufacturing and supply of products and related services in demand.
IRIS	Data Management Platforms, Artificial Intelligence-based Decision Support Systems, and Digital Platforms.
ALFA	The project will enable ALFA to understand where adoption of new technologies is of best value to the business and it's skilled workforce
KNEIA	Will extend its vision to build national and international cooperation networks with the aim to create new prospects for business development
WEB	The introduction of new technologies will enable WEB to expand its project range, adapting burners for the future growing marker of hydrogen burners as industry moves towards decarbonising fuel sources and processes
FORD	The project will deliver key improvements in the partner's capabilities in extending the digital thread from the engineering phase of a project through to the installation and launch phases, whilst ensuring that pre-fabrication verification work is robustly translated into the physical environment in full consideration of both technological and social factors.

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SCALE AND SIGNIFICANCE OF THE PROJECT´S CONTRIBUTION TO THE EXPECTED OUTCOMES & IMPACTS

The categorisation defined above applies to individual job roles; a company with many roles may well identify, that different roles fall within different categories.

Within the scope of the Up-Skill project, following on from the categorisation by class leading to defining the approach (the How) we will also quantify the impact of adoption of the industry 5.0 approach defined in terms of benefits from implementation, including:

1) Cost savings – reduced costs in primary manufacture.
2) Quality of output – addressing reproducibility vs. artisan skill as appropriate to define levels of improved productivity within specification, considering the human factors which define that specification for a given application.
3) Improved productivity – addressing both productivity within current production methods and also the reduced development cycle for new products.
4) Waste reduction – both by enabling higher levels of reproducibility where required and by defining specification tolerances with better fit-up between components through software-matching of parts, enabling use of parts which would previously have been out of tolerance.
5) Longevity of demand for skilled labour/craft skills – through enabling better quantification of value-add through the development of a reference framework which can be used to express value across the business, to both the craftsman and the management levels, enabling training and recruitment decisions to be business driven, converting the universal 'them and us' perception of workers as a wage problem to one where the value-add to the business is understood at all levels.

		Expected Impact					
Industrial use case	Outcome category	Cost savings	Increased sales/ revenue	Quality of output	Improved productivity	Waste reduction	Longevity of demand
Manufacturing and Engineering	AAA	□□□□	□□	□□	□□□□	□□□	□□
	AHH	□□□□	□□	□□□□	□□	□□□□	□□□
	HAA	□□□	□□□	□□□□	□□□	□□□	□□□□
	HHA	□□□	□□□□	□□	□□□□	□□	□□□□
Automotive industry	AAA	□□□□	□□□	□□	□□□□	□□	n/a
	AHH	□□□□	□□□	□□	□□□	□□	n/a
	HAA	□□□	□□□	□□	□□□□	□□	□□
	HHA	n/a	n/a	n/a	n/a	n/a	n/a
Process Industry	AAA	□□□	□□□	□□	□□□	□□	□□
	AHH	□□□	□□□	□□	□□□	□□	□□
	HAA	□□□	□□□□	□□□	□□□	□□	□□□
	HHA	□□□	□□□	□□□	□□□	□□	□□□
	AAA	□□□□	□□□	□□	□□□	□□□□	□□
	AHH	□□□	□□□	□□	□□□□	□□□	□□

Electronics and electrical systems	HAA	□□	□□□	□□	□□□	□□	□□
	HHA	□□	□□□	□□	□□□	□□	□□
Information and communications	AAA	□□□□	n/a	n/a	□□	□□□	n/a
	AHH	□□	n/a	n/a	□□□	□□	n/a
	HAA	□□□	n/a	n/a	□□	□□	n/a
	HHA	□□□	n/a	n/a	□□	□□	n/a

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Project outputs and expected impacts

Project outputs	Expected impacts
Cross-fertilisation of latest developments in implementation of technology for I4.0 with best practice in sociological research.	Recommended strategies for implementation make use of the best available data, increasing impact.
Coverage of applicable range of workplace scenarios	Broadest possible range of beneficiaries
Detailed understanding of human factors within craft skilled roles	Categorisation framework against which to define skill value and consequent approach (substitution vs. augmentation)
Example case studies	Establishes a gateway for engagement between the consortium and industry in delivery of consultancy-based support beyond the project.

Best practice technology solutions selected	Most effective solutions implemented
Enabling verification of identified approach	High level of confidence in recommendations to industry and government
Validation of approach	Value-add to case studies
Case studies demonstrating the impact of implementation	Enables self-assessment by industry across the UK, through identification of best fit examples and evaluation of impact of adopting particular strategies.
Enabling validation of identified approach	High level of confidence in recommendations to industry and government

Future skill needs identified	Informs government industry and training providers in developing training programs in advance of future needs, ensuring skills are brought on line without delays

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Potential Challenges and Obstacles	Odds	Impact	Contingency Measures and Conditions
Cost barriers: early adopters of industry 4.0 are alreadymaking significant investments in unproven technologies, while majority still holds back, waiting for standardization.	High	High	I4.0 has great potential and has so far garnered significant traction across board, - €140B annually planned investment for the European industry for the next 5 years, to support transformation in the manufacturing, engineering, automotive, process industries, …
Skilled experts: most manufacturers lack skilled personnel that fully understand how to leverage industry data appropriately, slowing down the full implementation of industry 4.0.	Med	High	We will break down the barriers, learn about the technology, while hiring knowledgeable personnel and observing each case study (AAA; AHH; HAA and HHA) to ensure smooth rollout.
Resistance from management: there is a low level of awareness about I4.0 and its benefits including from top management which has led to widespread resistance to accept the change.	Med	Med	We will conduct extensive outreach initiatives to industry stakeholders to assuage labour concerns and deliver the message that our technology is designed to pair skilled craftsmen and machines.
Social barriers and/or requirements: Solutions too complex to be used effectively.	Med	Med	We will prioritise ease of use to increase the adoption rate of the service within our target stakeholder groups. The new features will be user-driven: users selected from demonstration sites and throughout their current network will regularly be consulted and their feedback used for developing product iterations.
Regulatory barriers and/or requirements: unclear legal situation concerning the use of external data and unresolved questions concerning data security including lack of standard and regulations.	Med	High	Data privacy is a priority for every customer and sometimes they prefer the data storage on premises vs. in the 'non-EU' cloud. We will offer assurance that the cloud is in a certain EU location or from a certain provider, according to client preference.

2.2. MEASURES TO MAXIMISE IMPACT

DISSEMINATION PLAN

The consortium fully supports the view that a dissemination and exploitation plan is essential to maximising impact, through creating awareness of the project results and achieving the future exploitation of the project. In recognition of this importance, two WPs (6 and 7) are dedicated to these actions. This plan will integrate along the project many activities to enhance the dissemination and exploitation strategy, maximize the expected impact and boost the project's sustainability for its continuation beyond the EU-funding period.

Dissemination and exploitation will mainly focus on (i) Creating awareness of the project potential through dissemination of the results obtained from the WPs; (ii) Performing dissemination of success stories related to the innovative developments within the project; (iii) Making the scientific community, stakeholders and decision makers aware of the potential uses of the innovative approach to assessment of the value of human capital developed in the project; (iv) Convincing end users about the added value of the technology; (v) Collecting feedback from the future

end users and early adopters to adjust the exploitation plans; (vi) Creating the basis for the exploitation of the results that will be obtained.

In order to structure the whole plan for these actions, the deliverable “Plan for Dissemination and Exploitation of Results” (PEDR) will be created Month 6 as part of the WP7. This will integrate in schedule described in the table below.

Scheduling of dissemination plan timing

	Within project lifetime		Beyond project lifetime	
Dissemination needs	Initial awareness phase (M1-M12): Create the project visual identity, develop the main dissemination tools and identify some initial dissemination opportunities	Targeted awareness phase (M13-M24): Dissemination activities that will increase awareness about Up-Skill technologies	Full awareness phase (M25-M36): Dissemination activities that will keep end users informed and possibly increase the market uptake	Awareness after project completion: Dissemination activities that will promote the project for further exploitation
Dissemination objectives	• Raise awareness • Inform about Up-Skill actions to stimulate the interest of the full range of potential end users to ensure maximum exploitation of the developed foregrounds	• Increase and maintain awareness (including objectives & expected impact) • Keep interested potential end users informed on the latest developments of the project	•Increase intensity of awareness by having more promotional activities to target end users •Lobby with policy makers and public influencers about potential of Up-Skill technology and its impact on the public and environment	•Commercial exploitation •Incorporation of Up-Skill technologies into best practices for the relevant industry
Communication tools & measures	Project website, project logo, posters, leaflets, presentations and poster templates	Update PEDR if necessary Identify target audience Keep Project Website up to date with new developments	•Update PEDR if necessary •Identify target audience •Keep Project Website up to date	•Ensure project website is alive and updated even after project completion • Continuation of promotional videos

	• Use of social media (LinkedIn, Twitter, etc) • Participation and organization of conferences and/or workshops • Develop the Plan for the Exploitation and Dissemination of project Results (PEDR) based on Support Services for Exploitation of Research Results (SSERR) recommendations	• workshops, events, and conferences • Press releases and newsletters • Advertisements and promotional films on social media • Cross-dissemination with the activities of other organizations • Open access publications according to requirements (8 publications) • Presentations at conferences (at least 10)	• With new developments, workshops, events, and conferences • Press releases and newsletters • Advertisements and promotional films on social media • Cross-dissemination with the activities of other • organizations • Site tours to relevant facilities	• Materials in social media • Continue site tours to relevant facilities with the combination of workshops

The scheduling above will be integrated around three main pillars, focusing on awareness, understanding and

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action. [1] Dissemination for Awareness: to ensure Up-Skill is known to relevant stakeholders, development funds, municipalities and public authorities' planners, and the public. Dissemination for Understanding: to encourage a better understanding of the project results leading to wider engagement of external stakeholders with the project, and a higher potential for future uptake of the project outcomes. Dissemination for Action: To ensure adoption of technologies, processes and services being developed by the project partners.

For the dissemination of the results and information on the project activities, the plan will contain the following information, including details on the key project results to disseminate:

- Definition of the target groups (policy makers, industries, R&D stakeholders). Besides th stakeholders, the dissemination of results will also reach the general public via the communication actions described below, which also include institutional stakeholders.
- Definition of the message (i.e., enhancing the role of digitalisation and demonstrating the positive impact of the project solutions, showing the development of the project, etc.).
- Selection of the dissemination channels (journal articles e.g., Journal of Manufacturing Systems, publications, conferences, exhibitions, website, emails, etc.).
- Planning of the implementation (timing, costs and human resources needed, etc.)
- Monitoring and development of Key Performance Indicators (KPIs) to assess the success of the implementation (number of publications, number of emails received from stakeholders, number of participations during project events, feedback received from audiences at conferences, surveys, etc.)

The table below outlines a presentation of the different type of user groups related to Up-Skill who will be targeted via all the dissemination actions and networking/clustering activities of the project. During the first stages of the project, a stakeholders' analysis will be executed to successfully identify the project end-user community and networking actions required.

Definition of target groups and dissemination objectives

Targets and user groups	Description and dissemination objectives	Objectives	Dissemination user groups dissemination objectives content and channels

Academic communities, research communities	This group targets all research communities interested in the project's deliverables, research developments, results and innovation scientific community which can be beneficiary for their research activities. Innovative contributions of the project are particularly interesting for researchers.	Transfer of knowledge, raise awareness	Public deliverables, scientific publications, conferences and other scientific events.
Industrial sector	An important objective of Up-Skill is to address and trigger the active involvement of the industrial sector. TWI is a membership organisation with 600+ industrial members (aerospace, oil & gas marine, heavy industry, automotive, electronics, …) across 40 countries who benefit from TWI 70 years of knowledge and expertise in research, contract R&D, technical, standards information and engineering services. TWI actively participates in 100+ national and international standards committees.	Demonstrate the business potential of inclusivity towards early adoption of products and services developed by the consortium, collect feedback on their expectations and requirement to adjust commercial exploitation plans, convince about the technical feasibility and competitiveness of the concept and tools developed.	• Market test results, techno- Economic assessment, etc. • Dedicated workshops, Public deliverables, Scientific publications, related project Events.

Government bodies and policy makers	This is a wide group encompassing innovation driven local and regional authorities, representatives and associations, Ministries, parliaments and Public Administrations at national and international level.	Demonstrate the benefits of Up-Skill concept and tools to reach the inclusive European future goals, raise awareness about proposed regulatory evolution.	Final recommendation in deliverables, press kit, general dissemination, participation in policy events
European and international technology Networks	This group refers to activities addressing external task forces.	Use as dissemination relays towards their members	Public deliverables, press kit, articles, press releases, communication Package
EU projects working in similar domain	The participation of project partners in other relevant projects offers the opportunity to establish quick links among parties through joint actions. A list of these has already been established, as presented in Section 1.2, with a more detailed explanation on why to reach to them.	Coordinate dissemination activities in order to maximize their impact, exchange R&D results to improve robustness of project results.	Dissemination events, presentation at conferences (TWI Digital Manufacturing), participation in workshops from other projects.

The general public	The general public consists of a general audience and other actors not identified as direct targeted groups by the project, though this group can have strong interest in the project.	Raise awareness on the importance of inclusive future work and decent life for EU citizens specifically as a credible alternative for the future, to inform about the benefits of the project towards a sustainable production system.	Project website, brochure, press releases, social media, project events

At the end of the project, we plan to elaborate the dissemination impact analysis where we will evaluate the responses gained from the different target groups. This will bring important information for further exploitation of the project results by particular consortium partners after the end of the project. In addition to the external action, within the project there will also be the creation of an External Advisory Board (EAB) to achieve broader industrial engagement. This board will provide valuable feedback on the project, introduce challenging requirements to be considered and have a major impact on the project's sustainable development. The EAB members will also participate in the dissemination actions.

Quantifieddissemination monitoring (KPIs)

(a) The partners will participate in at least 10 conferences where the project results will be presented either via oral/poster presentations and where this could lead to the publication of conference proceedings. Darren is highly involved in TWI Digital Manufacturing Conference, an innovation network that works to nature and coordinate partnerships that enable engineering advances in specialist areas in response to commercial needs. In addition, TWI is actively involved in NSIRC Annual Conference – a showcase of industry relevant PhD research projects, including additive manufacturing, robotics and sensors, attended by experts from industry interested to utilise the research in their R&D programmes.

(b) The partner will make 10 publications in conference proceeeding

(c) The partners will publish at least 8 scientific publications in peer-reviewed journals, for example the Journal of Manufacturing Systems and the Journal of Manufacturing Technology Management.

(d) The partner will make publication through websites, social web-based media, promotional videos

(e) The partner will participate in at least 3 industry conferences

(f) The partner will participate in at least 4 academic conferences (minimum two per year in the last two

years)

(g) The partner will organize at least 2 international technology workshop and 1 final information day

(h) The partner will deliver webinars through regional and national “Industry 4.0” promoting bodies. Minimum of 9 (3 each in Italy, UK and Sweden)

(i) The partner will involve at least 150 interested stakeholders in the project in each of the participating countries (450 firms made aware of the project’s work) through direct contact, webinars, and other presentations.

(j) The partner will participate in 6 national trade fairs (2 in Italy, UK, Sweden)

(k) The partner will report on cost analysis and positioning of technology

(l) TWI will participate as exhibitor in at least 3 industry conferences to promote the project results

(m) In addition, the consortium will organize one technology workshop related to the project results with a target of 40 interested stakeholders.

EXPLOITATION PLAN

Creating markets from research results is becoming a requirement to boost research, towards a constant evolution in which universities and research centres are engaging with companies and the non-academic and/or private sector. By its nature, Up-Skill holds the potential for exploiting research results at various levels, involving different types of organizations, such as industries, research centres as well as other public interest organizations, for the benefit of the affected stakeholders. The plan for exploitation, which features activities in WP5&6, will consider these different levels and will present the different aspects below:

- Target groups identification: lead and affected stakeholders principally. The range of organization types targeted by Up-Skill is wide enough to ensure an efficient outreach, while, at the same time, protecting project results vis a vis external developer and preserving the interest of all partners.
- Lists of outputs to be exploited (Table 9) and explanation on how to proceed with their development.
- Definition of the exploitation and valorisation strategy (improvement of dissemination strategies, market and needs analysis, feasibility studies, research of funding, standardization, commercialization and transfer of knowledge, ...). This strategy will outline potential trends, technology scenario, partners and competitors in the areas of interest for the project This will also include the implementation of performance indicators to assess the adequate valorisation strategy.
- Methods of exploitation (patents, company creation, private public partnerships, …). The conclusion of a fair and generally agreed Consortium Agreement before the start of the project represents an important first step towards a sound exploitation of the project results. Then, the range of different possibilities for exploitation should be studied and analysed towards the TRL of the technologies provided, in order to develop the appropriate plan, which takes also into account the nature of the partner(s) involved in the dissemination and exploitation and of their financial conditions.

This exploitation plan will be an evolving document with its initial form presented here, and it will be updated periodically during the course of the project. Concrete measures for exploitation have already been identified by the consortium according to the exploitable results of the project and will be further gathered in the final exploitation plan to be delivered Month 6. The major project outputs already identified and that will have the most value for exploitation:

List of exploitable outputs

Types of results	Owner	Exploitation route and protection	Potential users	Dissemination to ensure exploitation
Creation of scientific data	All	Open databases	Basic and applied researchers, Engineers, Developers	Dissemination in open access, Lectures and courses, Publications through websites and Journals.
Reports & Deliverables	All	Through the publication of scientific papers	Basic/applied Researchers	Participation in scientific conferences, participation in trade Fairs
Report on cost analysis and positioning of the technology	All	Participation and support to EU policies	EU and national Policy makers	Dissemination to policy makers through EU networks and EU associations

Actions planned to achieve the exploitation of the results and the impact of the project

Types of action	Description	Targeted groups
During the project		
Events	Organization of 1 international technology workshop addressing the main technology application of the project.	R&D sector, the academic and non-academic organizations with specific players in the field, the public at large, EU projects
		interest.
Events	Organization of 1 final info day: The impact and further exploitation of the project results will be introduced to a wider public.	Public at large, policy makers, media, and all other stakeholders
Internal event	Exploitation and IPR workshop	Consortium
Internal event	Participation to IPR webinars organized by the IPR EC help desk.	Consortium (at least one webinar during the course of the project).
Recruitment of early-stage researchers.	Recruitment of at least 4 PhD students and 6 pos-doctoral positions who should moreover be especially relevant candidates for the creation of innovative start-ups.	The PhD students and post-docs themselves Special emphasis will be put on training them for innovation creation.
IP Protection	The creation of patents is planned to guarantee the good protection of the project results to ensure a good exploitation management.	Direct stakeholders for end-use.
Market test assessment	The realization of a market test with the outreach to more than 80 stakeholders will boost further engagement into the project.	End users of the technology, policy makers.
Work package tasks	The implementation of a WP dedicated to the scale up of the project solutions. The objective of this task is to provide useful information to Public Policies at larger scale about the interest and chance of success to scale up the project solutions, where it's the most judicious to introduce it for the larger expansion.	Policy makers and industries
Collaboration with other projects	Results might be useful for fundamental information, system evaluations, innovative inputs, network publication.	Stakeholders from national, EU and international funded projects.
Creation of external advisory board	To ensure management of innovation within the consortium and guarantee that all measures are taken to maximise dissemination and potential exploitation of the results during the project. The board will include key actors in target areas and will be used as a tool to advise on related issues such as public acceptance, local regulations, fair inventive schemes and exploitation of project results.	Consortium actors in the development of key exploitable results.

Dissemination and communication	A strong dissemination and communication strategy is planned to ensure that strategic feedback on the technology is received and that dialogue between the consortium and stakeholders is reinforced to bring coherence and strategy to future development of the technology.	All audience.
After the project		

Research development	Future internal research at the partners' institutions will be carried out to ensure the further competitiveness of the technology development/upgrade. The results will be used as a background for future collaborative innovation projects. This should take place potentially through new EU funding or national funding.	ocument Ref. Ares(2022)4648166 - ortium partners new key actors in the field, especially targeting the industry actors
Creation of new services and products	The new services and products likely to be exploited by the SMEs of the consortium to propose new services or with the creation of spin-off and start-up companies.	SMEs, external industry actors
Further funding	Horizon Europe funding for innovation (especially the SMES instrument), public procurement, venture capitals, private investors, banks, business angels will be sought for, to ensure the further development of the technology.	End users of the project, investors
Roadmap to commercialisation	A wide dissemination of the roadmap will ensure that stakeholders engage with the project results even after the end of the project.	Industrial and policy makers
Open databases	The permanent access to the main results of the project in different databases.	All interested stakeholders

The development of the complete and consolidated business plan will be executed in the framework of WP6 and will take place in three distinct steps as identified below:

1) Feasibility Study: This will be the first step towards the definition of the business plan with the aim of finding out the workability and cost effectiveness of the business cases targeting the EU set-plan objectives. This will include calculations, examination and projected estimations of the business opportunity to identify the actual potentials towards adoption and commercialization of the technology. The project is developing understanding that will be commercialized through consultancy and indirectly through enhancement to future technology projects by enabling more effective uptake through due consideration of implementation approaches. The feasibility study will be based on technical, commercial, financial and organizational aspects including the description of the Up-Skill outcomes to be commercialized, the identification and analysis of the different sector markets and domains, financial assumptions regarding implementation, funding requirements, as well as personnel and organizational requirements. Following the success of this step, which proves the actual potential for the commercialization, the consortium will proceed to the next step towards the definition of the business modelling and the business plan development.
2) Business Modelling: This will be based on the utilisation of a business model canvas approach as a visual chart which defines the value proposition, infrastructure, distribution channels, key resources and activities and impact of the business outcome to develop the business models towards the exploitation of the Up-Skill solutions. The business modelling will also ensure the proper positioning of the solutions and use through the elaboration of an effective business cycle involving all project partners and stakeholders.
3) Business Plan: Following the above two steps, a complete business plan will be developed that defines the way that the project outcomes will be brought to market including steps towards and after the project end. It will also include identification of funding opportunities for continuing the project activities including private funds, funding through other national and/or regional R&I programmes, ... The focus of the business plan will be to maximize the impact of the project outcomes that follow:
   (a) Propose means to facilitate an increase of the market-readiness
   (b) Identify mechanisms to lower the entry barriers of the developed outputs into the EU and other markets.
   (c) Positioning the Up-Skill solutions as a means of contributing to the growth of competitive inclusive digitalisation in Europe.

COMMUNICATION ACTIVITIES

The communication activities that will be part of the dissemination plan will be tailored in order to ensure that important messages are widespread to the adequate targeted audience and that the public at large gets connected with Up-Skill. Such activities complement the dissemination as they “translate” the sometimes- complex results into easy-to-understand media resources focusing more on the impacts and added value for the endusers and society. The project will have an open access approach in several ways. The exploitation plan describes that the project will establish "all open databases" and execute an "open access dissemination".

The main purposes of the communication activities of the project have been defined as follow:

- To show how European collaboration has achieved more than would have otherwise been possible, notably in achieving scientific excellence, contributing to competitiveness and solving societal challenges;
- To show how the outcomes are relevant to people's everyday lives, by creating jobs, introducing novel technologies, or making people's lives more comfortable in other ways.
- To make better use of the results, by making sure they are taken up by decision-makers to influence policy-making and by industry and the scientific community to ensure a follow-up of the development of the technology.
- During the first stages of the project a stakeholder analysis will be executed.
- The members of the external advisory board (EAB) will participate in the dissemination actions.
- A dissemination impact analysis will be executed and elaborated in the end of the project based on the responses gained from the target groups.
- To show how European collaboration has achieved scientific excellence, contributing to competitiveness and solving societal challenges.
- To show how the outcomes are relevant to people's everyday lives, by creating jobs and introducing novel technologies.
- To make sure that the project results are adopted by decision-makers to influence policy-making and by industry and the scientific community to ensure a follow-up of the development of the technology.

The communication strategy will be led by KNEIA and conducted around seven main elements, discussed below. These combine a set of mass media communication (Press releases, website, brochures) and interpersonal communication (workshops, open days, conferences, etc) in order to reach the public at large as maximum as possible. These activities should be considered as non-exhaustive and will be updated as the project evolves. For each communication activity, a synthetic action plan will be inserted in the D&E plan to detail the methods planned to guarantee the success of the action (target groups, message to be conveyed, review procedures, impact monitoring, etc).

Visual Identity	The project branding will help all partners communicate about the project in a uniform, consistent, and professional manner. The project branding will include project logo, visual identity, templates for word and PowerPoint, written identity including tagline and key messages
Communication materials	A communication package containing the main elements of the project and gathered in a PowerPoint presentation, a poster and a Word document (one page project description, objectives, impacts) 1 flyer, 1 brochure, 8 newsletters (1 every 6 months), 2 press releases and 2 articles in specialized magazines, 1 timeline infographic, 1 motion design video to be promoted through the EU audio-visual channels and YouTube plus another one on the filming of the pilot plant in action Liaise with the different partners communication departments for wider dissemination of these materials using the partners existing communication channels
Website	The public website will contain information targeted for the general public (description of the project, the WPs, the partners, basic information on the technology). There will also be specific information targeted towards the different type of stakeholders linked to the project (training materials, scientific papers, environmental impact, …).
Social networks and online presence	Social web-based media (creation of 1 LinkedIn page and 1 Twitter project accounts, and which will service to target the general audience as well as more technology related stakeholders.

KPIs for communication in the UpSkill application:

(a) Presentation material: a PowerPoint, a poster and a Word document.

(b) 2 press releases per year in years 2 and 3 (4 total)

(c) 2 articles in specialized magazines per year in years 2 and 3 (4 in total)

(d) 8 newsletters, (I every 6 months)

(e) 3 flyers outlining different aspects of the project (technology developments, future of work, Industrious Modernity)

(f) 1 brochure (updated each year – 3 versions in total translated into English, Swedish and Italian) (g) 1 timeline infographic,

(h) 2 motion design videos to promote the project and present the pilot plant in action.

(i) Interviews of each of the partners will be done and integrated into a video for online distribution

(j) A website will contain information targeted for the general public and support tools developed by the project.

(k) 1 LinkedIn page and participation to the H2020 CSP projects twitter account.

In addition, there are also suggestions for activities beyond the project:

(a) Roadmap to commercialization: A wide dissemination of the roadmap will ensure that stakeholders engage with the project results even after the end of the project.

(b) Open databases, the permanent access to the main results of the project in different databases. All interested stakeholders

(c) The results will be used for future collaborative innovation projects to ensure the further competitiveness of the technology development.

(d) The new services and products likely to be exploited by the SMEs of the consortium to propose new services or with the creation of spin-off and start-up companies.

MANAGEMENT OF INTELLECTUAL PROPERTY

(a) Intellectual property rights

For the success of the Up-Skill project, it is essential that all project partners agree on explicit rules concerning IP ownership, access rights to any Background and Foreground IP for the execution of the project and the protection of intellectual property rights (IPRs) and confidential information before the project starts. Therefore, such issues will be addressed in detail within the Consortium Agreement between all project partners. The main purpose of the Consortium Agreement is to establish a legal framework for the project in order to minimize any internal issues within the UP-SKILL consortium related to the work, IP–Ownership, Access Rights to Background and Foreground IP for the duration of the project and any other matters of the consortium's interest.

(b) Access Rights to Background and Foreground IP during the project

In order to ensure a smooth execution of the project, the project partners agree to grant royalty–free access to background and foreground IP for the execution of the project. Therefore, all project partners determine any background IP they are willing to submit to the project within the Consortium Agreement before the project starts. Any details concerning the access rights to background and foreground IP for the duration of the project will be defined

in the Consortium Agreement.

(c) IP Ownership

Results shall be owned by the project partner carrying out the work leading to such results. If any result is created jointly by at least two project partners and it is not possible to distinguish between the contributions of each of the project partners, such work will be jointly owned by the contributing project partners. The same shall apply if, in the course of carrying out work on the project, an invention is made having two or more contributing parties contributing to it, and it is not possible to separate the individual contributions. Such joint inventions and all related patent applications and patents shall be jointly owned by the contributing parties. Details concerning jointly owned results, joint inventions and joint patent applications will be addressed in the Consortium Agreement.

(d) Consortium Agreement

The purpose of the Consortium Agreement is to establish a legal framework for the project in order to minimize any internal issues within the UP-SKILL consortium related to the work, IP Ownership, Confidential Information, Access Rights to Background and Foreground IP for the duration of the project and any other matters of the consortium's interest.

2.3. SUMMARY

KEY ELEMENTS OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Manufacturers seek solutions that will guide the integration of craftsmen and machines to fully extract benefits that can be reaped from it. In the near future, every data consumer company will be demanded to have experts dedicated to take decisions based	i. Advancing the knowledge on future skills requirement and contributing to the systematic investigation of working practices associated with craftsmanship in contexts characterised by automated manufacture of some parts. ii. Assessing how Italian policies influence employment, skills acquisition and retention, and industrial competitiveness iii. Offering empirical knowledge on Industrial design sector and advancing the understanding of the current state of skills needs and training within the sector in Italy iv. Two Post-doctoral Researchers based in Milan to carry out fieldwork, review the literature	Communication with citizens - will comprise a range of activities, including publications; presentations; the use of social media networks and on-line platform websites; audio-visual media (e.g., podcasts), Dissemination - To maximise the impact of the research, actions will be directed to: 1. The academic community; 2. Non-Scientific audience: (a) policy- makers involved in designing employment policies to foster industrial competitiveness (b) National and EU organisations: to create awareness of the importance of the topic among organizations interested in promoting employment and training options at both the local and EU level; 3. Wider industry: craft businesses, manufacturing businesses reliant on skilled workers and

on validated information.	and analyse cases, will be recruited.	organizations promoting craft and manual skills at national and local levels.

TARGET GROUPS	OUTCOMES	IMPACTS Associated with document Ref. Ares(2022)46 24/06/2022
Manufacturin Industry Artisans	• Pathway for combi and craft skill, eith designated role or skilled role embed manufacturing • Validated recom	Scientific: Establishing a new methodology for reviewing, interpreting and describing the emerging field of human/technology interfaces created by I4.0 Economic:
Cultural Preservation	for I5.0 integration • Establishin skills and manufactu • Assessmen routes access ben	• Enabling ongoing manufacturing of products where skills are at risk, estimated to retain 15,000 direct jobs and underpin 200,000 jobs across the EU • Improving the utilisation, efficiency of highly skilled workers, estimated to increase productivity by 5% on average Preserving innovation capabilities Societal: • Establishing training programmes to meet identified future needs • Enhancing and growing opportunities for decent work Cultural: • Preserves traditions and cultural heritage • Increased ease of repair and retention of cultural heritage

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

Associated with document Ref. Ares(2022)4648166 - 24/06/2022

3.1. WORK PLAN AND RESOURCES

OVERALL STRUCTURE OF THE WORK PLAN

The project is organised in seven main work packages as shown in the WP breakdown (Figure 7).

WP1 - Enabling Technology Assessment
To carry out a systematic gap analysis for interoperable solutions through the expert review of emerging enabling technologies for I5.0 with particular focus on the interfaces between them. Capturing a wide portfolio of studies and tools will build system capability for use in WP2 and the web based Up-Skill platform, which is the public dissemination and system tool approach.
WP2 - Data collection (use cases)
Development of case study requirements within manufacturing/production businesses, reviewing skilled roles, customs, habits and cultural difference in their approach and identifying SWOT for I5.0 within them. The generated data (per each case) feeds in to WP3 and the Up-Skill platform. The Malardalen region, Sweden and Milan region, Italy will be studied as regional ecosystems.
WP3 - Use Case Development: Skills vs Quantified Products

Development of case studies within supply chain settings, reviewing 'hand built' product with respect to opportunities for data-driven improvements in yield/ultimate performance. Matching equipment and ethnography, requirements with WP4 to augment Up-Skilling 'hand built' processes, boosting quality, output and job protection, the methods and results being captured in the Up-Skill platform.
WP4 - Training needs assessment

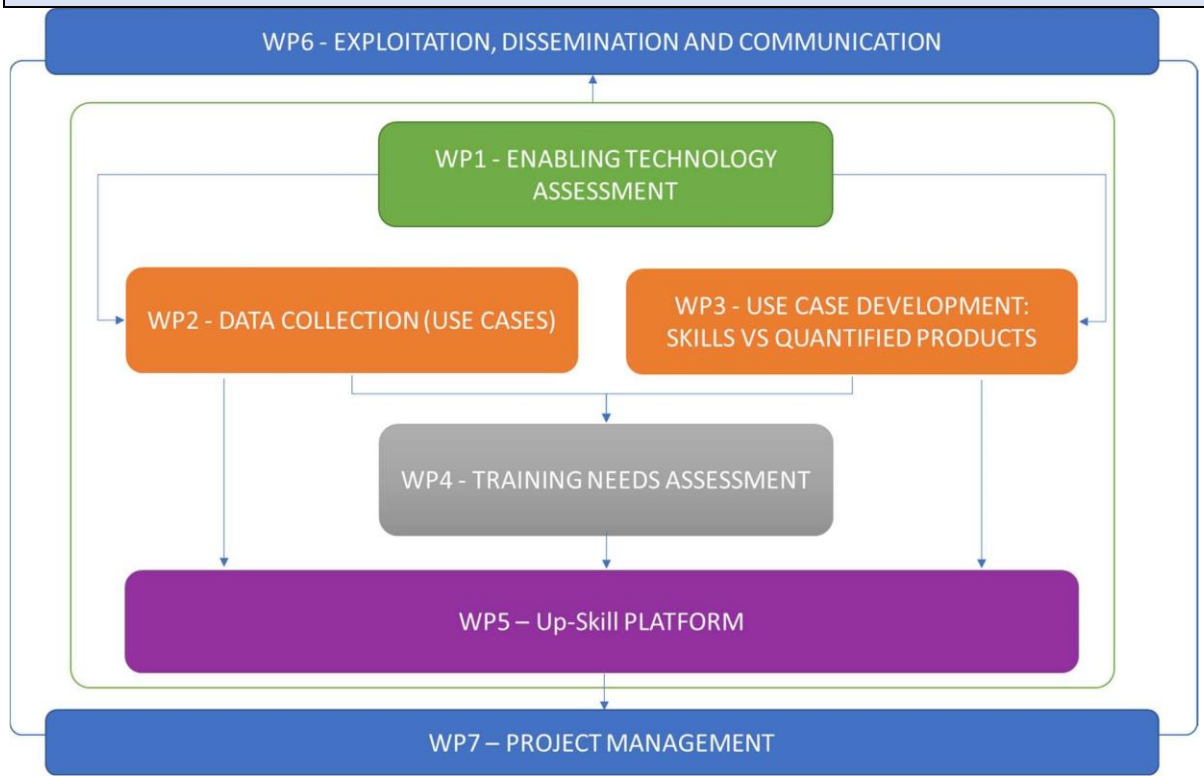

Figure 7: Up-Skill Pert Chart – Overall Structure

This WP extracts the training needs from each use-case, in conjunction with the results acquired from the previous WPs, enabling a comprehensive set and widely accepted user requirements that translate to the Up-Skill platform (WP5) for use by industry. The case studies also benefit from tailored training and programming to meet their particular manufacturing conditions, for use and with pertinence to broader industry through the Up- Skill platform.
WP5 – Up-Skill Platform
This WP draws all of the data and findings from WP1, 2, 3, and 4 together, and packages them into our intelligent Up-Skill platform where all stakeholders can engage in industry 5.0 adoption from SMEs to large enterprises, to policy-makers and individuals. Up-Skill will host all of the case study findings, training programmes and future skills mapping across all industries.
WP6 - Exploitation, Dissemination & Communication

Associated document Ref. Ares(2022)4648166 - 24/06/2022

The goal of WP6 is the dissemination of the project to a wide audience, conveying the Up-Skill concept, development case studies and their findings to key actors, and integrating their feedback. Additionally, exploitation plans for key project results will be derived including video outputs and the web-based platform to extoll best practice and likely development pathways to achieve holistic company/staff benefit. Therefore Up-Skill's business delivery methodologies will be presented as a learning-based approach.
WP7 - Project Management
Project management will ensure the achievement of the project objectives within time and budget constraints, by planning, organizing and controlling the integrated effort of the consortium. This will be a continuous task throughout the project. Consortium partners belong to different fields, organizations, enterprises and effective information flow/communication is necessary due to heterogeneous disciplines. Management will be realised by: a) Build-up/support knowledge, and experience exchange, b) Meetings with all project partners and if necessary, visits of the coordinator to partner sites, c) Resolving problems if they appear. Detailed explanations on the management structure, decision-making, communication methods and instruments, quality assurance, instruments for monitoring and reporting progress and management of knowledge and intellectual property are given in the section "Management Structure and Procedures".

UP-SKILL GANTT CHART

WORK PACKAGES		GANTT																																			
TASK DESCRIPTIONS	LEAD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
WP1. ENABLING TECHNOLOGY ASSESSMENT																																					
T1.1 (Design) Review of ethnographic approaches, and method planning with use case development	ARU																																				
T1.2 Ethnography workshop review	ARU																																				
T1.3 Literatuire review / technology gap identification	TWI																																				
WP2. DATA COLLECTION (USE CASES)	ARU																																				
T2.1 Case study assesments	ARU																																				
T2.2. Ethnography research period	ARU																																				
T2.3 Detailed case study write up for Up-Skill platform	ARU																																				
WP3. USE CASE DEVELOPMENT: Skills vs Quantified Products	LUN																																				
T3.1 Identification of enabling technologies	LUN																																				
T3.2 Manufacturing workflows	LUN																																				
T3.3 Hardware identification and instillation	TWI																																				
T3.4 Trial Manufacture (with ethnography)	TWI																																				
T3.5 Trial feedback back case study	LUN																																				
WP4. TRAINING NEEDS ASSESSMENT	MITC																																				
T4.1 Digital outcomes and training requirements	LUN																																				
T4.2 Programme training and usage methods	MITC																																				
T4.3 Hardware training	TWI																																				
T4.4 Evaluation of systems effectiveness	MITC																																				

T4.5 Future skills mapping	UMIL
WP5. UP-SKILL PLATFORM	IRIS
T5.1 Design and requirements definition for data capture and representation	IRIS
T5.2 Data repository development, data capture processes and pipelines, data population and formatting	IRIS
T5.3 Decision Support System (DSS) development using advanced data processing algorithms (AI techniques)	IRIS
T5.4 Validation testing on-line/in-situ with KPIs and integration with use cases	IRIS
WP6. EXPLOITATION, DISSEMINATION & COMMUNICATION	KNEIA
T6.1 Communication Activities	KNEIA
T6.2 Dissemination Activities	KNEIA
T6.3 Exploitation of results, IPR and Business model	KNEIA
T6.4 Clustering and Networking activities	KNEIA
T6.5 Stakeholder analysis and engagement	KNEIA
WP7. PROJECT MANAGEMENT	TWI
T7.1 Technical Management	TWI
T7.2 Financial Management	MDU
T7.3 Project Reporting	MDU

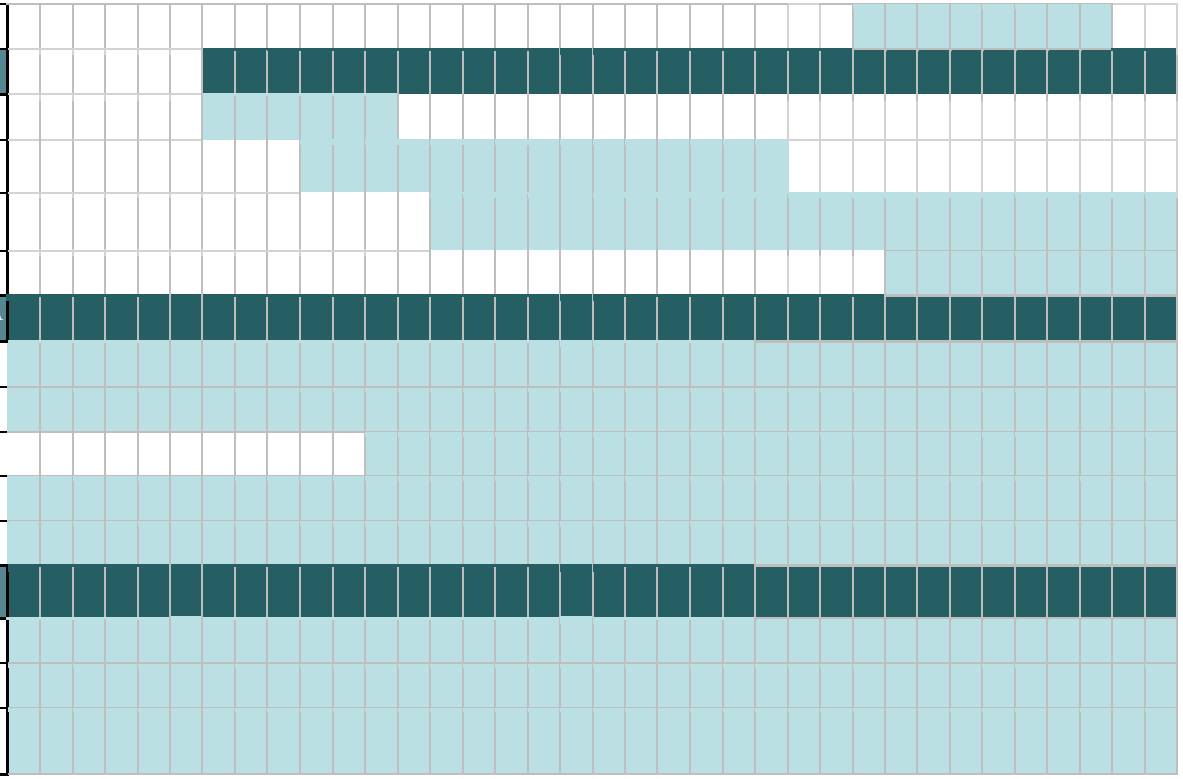

Table 3.1h: Purchase costs items (travel and subsistence, equipment and other goods, works and services)

Total		

Mälardalen University (MDU) is a progressive and collaborative university and one of Sweden's largest institutes of higher education. The University is characterised by its close partnerships with international companies in the public sector such as ABB, Volvo and Bombardier and HEIs all around the world. Our high investment in research generates beneficial solutions for societal development.	MDU will work closely with SSH partners ARU an UMIL in WP2 and will lead the research effort in the Malardalen region of Sweden, where the case studies of HEK, and Alfa Laval are situated.

7/KNEIA		
	Cost (€)	Justification
Other goods, works and services		
Remaining purchase costs (<15% of pers. Costs)		
Total	47,500	

8/WEB		
	Cost (€)	Justification

3.2. CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE

Mälardalen Industrial Technology Cent AB (MITC) seeks to develop industry, academia and society by transfering academic knowledge into activities of industrial companies. MITC work directly with the manufacturing industry, having member companies such as Volvo CE, Volvo GTO, GKN, Alfa Laval and Hexagon. MITC has worked with over 150 SMEs with the area of focus being manufacturing, industrialisation, and product development with digitalisation and sustainability as cross-sectional areas. Lately particular focus is being put on advanced robotics, cyber-physical systems and artificial intelligence. MITC has years of experience running coaching and development programmes with SMEs. Several effective programmes have been developed in areas such as digitalisation, Lean and continuous improvement and production development.	Mälardalen Industrial Technology Center will lead WP4 - Training needs assessment. MITC will build on their existing portfolio of training courses in and around the area of digitisation and sustainability, specifically aimed at SMEs. MITC will also provide additional case studies, which have been preliminarily confimed by GKN and Volvo GTO if any gaps are identified following undertaking the case studies at consortium partners.

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UNIVERSITÀ DEGLI STUDI DI MILANO The University of Milan (UMIL) is a leading institute in Italy and Europe for scientific productivity, being the largest university in the region and also an important resource for the socio-economic context of which the university is a part. With experienced researchers in the fields of organizational behaviour and organizational studies, there research focus is on the nature of contemporary workplaces and the ambivalence surrounding job transformation and alternative forms of work arrangements. UMIL have extensively researched the changing relationship between new organizational models and the quality, nature and meaning of work.	MIL will lead the research effort in the Mila Associated with document Ref. Ares(2022)4648166 - 2 region of Italy focussing on a range of Italian companies that are involved in the production of aesthetics and art goods, and that are characterised by a certain level of skilled craft in the production process. The focus is on small and medium scale companies involved in styling, designing and prototyping industrial design products in a diverse range of sectors including furniture and furnishings, interior design and yacht design, such as companies like Cassina, Edra, Anonimia Castelli, Benetti.
helekopter Helekopter AB is a company creating, developing and building company structure around SME-companies. Aiming to minimize administrative work, risk, and time to market. Sharing organizational resources and innovative best practice over different market segments and branches. Engage in how future education system can be adjusted to fit SME-companies need of students with broad understanding cross companies disciplines.	To assess, capture and verify the interface change between a SME-companies administrative work and business driven work, driven by customer new behavior and digitalization. Implementing the learning and create the changes for SME-companies continues growth with next generation employees
IRIS IRIS is a Deep-Tech Engineering company specialised in Real Time Monitoring and Industrial Artificial Intelligence Solutions for driving improved process quality and efficiency for industry. Our challenging and complex international projects provide our teams with the ideal environment for furthering their knowledge and upgrading their skills. Deploying AI and machine learning to connect upstream and downstream processes across the entire value chain. IRIS' work has resulted in 15 patents granted since its formation in 2007	IRIS will act as designer and developer of the digital tools as well as provider of AI algorithms associated with the DSS developed and deployed in WP5. The expectation is to fulfil the objectives of the WP successfully and contribute to achieving the global objectives in the project, Also, the development of a robust and commercial DSS and digital platform ready to be protected by IPR rights (patents, publications, ownership, etc...).

ALFA LAVAL (ALFA) is a leadi global provider of first-rate products in the areas of heat transfer, separation and fluid handling. With these as its base, Alfa Laval aims to help enhance the productivity and competitiveness of its customers in various industries throughout the world. We define their challenges and deliver sustainable products and solutions that meet their requirements – mainly in energy, the environment, food and the marine industry	ALFA will make up one of the case studies Associat with document Ref. Ares(2022)4648166 - 24/06/2022 investigated. ALFA undertakes a variety of engineering quite different activities (different markets, different volumes) falls under a number of categories. However, they are keen to advance their level of industry 4.0 capability and automate where they can.
KNEIA deploy a multidisciplinary approach in order to satisfy clients and partners in the fields of exploitation and dissemination consulting, innovation based scale up, training, plus technical direction and assistance. Our vision is to build national and international cooperation networks with the aim to create new prospects for business development in Spain and other EU countries. The group already have an existing relationship with IRIS through their partnership in the Horizon 2020 'PRESERVE' project.	KNEIA will lead the exploitation, dissemination and communication activities
Webber Brennertechnik GmbH (WEB) has been developing and supplying gas burners for industry, commercial devices and household appliances for 35 years .	WEB are a case study in Up-Skill looking at emerging technologies to solve a production problem. WEB can't offer specific burners to customers due to the only two specialist employees who were able to manufacture them unable to due to health reasons.

3.2 Associated Partners

Partner Name and Capacity	Role in the Project
TWI is one of Europe's largest independent, , membership-based research and technology organisations. TWI has a multidisciplinary team of consultants, engineers and technicians dedicated to the development and application of advanced inspection techniques, structural integrity assessment and risk management. TWI is internationally renowned for its collaborative research, technology projects and teams who can add value to almost every industry sector – from automotive, offshore oil and gas, civil engineering and nuclear power, to electronics, advanced aircraft (both civilian and military), medical equipment and solar power.	To assess, capture and implement the hardware requirements for the case studies. By agreement and liaison between the ethnography and the programming and by undertaking suitable training both for and on behalf of the case studies, partners and reporting the findings thereof.
Anglia Ruskin University (ARU) is one of the top 350 universities in the world and one of the top 40 universities in the UK. It is one of the institutions predicted to be challenging the global elite by 2030. The Faculty of Business and Law at ARU is a full-service business school and host to four research centres. One of these, the IMPact research centre led by Professor Chris Ivory, is already home to ERDF, Interreg and FORTE-funded (a national Swedish Funding body) projects on issue ranging from digitalization and Industry 4.0 to the green economy.	ARU will coordinate the ethnographic research effort over the life of the project (leading and organising research training, co-ordination meetings, knowledge exchange workshops and site visits). ARU will also lead on the writing of related cross-case study reports, the literature review, interim reports and final report.

Lancaster University is among the best in the UK., ranked amongst the country's top 20 universities. Lancaster has a world-class reputation for research and teaching excellence, with many of their academic leaders within their fields of study and their subjects consistently highly ranked in major league tables.	Associated with document Ref. Ares(2022)464 24/06/2022 Supporting ethnographic study researchers based at end user organisations Lancaster University will develop and apply a intelligence and machine learning across the us
Ford Motor Company (FORD) is a global company that designs, manufactures, markets and services a full line of Ford cars, trucks, SUVs, electrified vehicles and Lincoln luxury vehicles, and is pursuing leadership positions in electrified, autonomous vehicles and mobility solutions. Ford Powertrain Manufacturing Engineering (PTME), Europe is a department of Ford of Europe responsible for planning and installing production facilities across Europe for the manufacture of electric power units, transmissions, engines & castings for the Ford vehicle line up. Ford has launched multiple Industry 4.0 initiatives to support their business from the initial installation to the ongoing running of operations.	FORD have ideated an automated method utilising AGVs to undertake the floor outlining process generating data and eradicating the risks incorrect install location. Additionally, post-install verification would be undertaken using a laser scanner onboard the AGV to collect data to precisely cross-reference against the planned location. However, engineers are reluctant to move away from the current manual process due to a distinct lack of trust that the Ind 4.0 technology could be incorporated successfully. This case study, would look to leverage the skills of the consortium to develop a suitable AGV based system that could overcome adoption reluctance and lead to lessons that could subsequently be deployed to improve the human AGV interface across Ford. Therefore, this project is appropriately situated at the cross-section of Ind4.0 and social sciences.

4. Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact
The work that will be performed as part of Up-Skill project, will not include any clinical trials or involvement of children. The proposal is particularly focussing on mapping the skills and training needs of the future workforce with increasing adoption of disruptive emerging technologies in industry 5.0. We do not foresee any impact of the project's activities to particular social groups and we do not foresee any creation of political or financial adverse consequences.

Compliance with ethical principles and relevant legislations
No issues have been identified

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